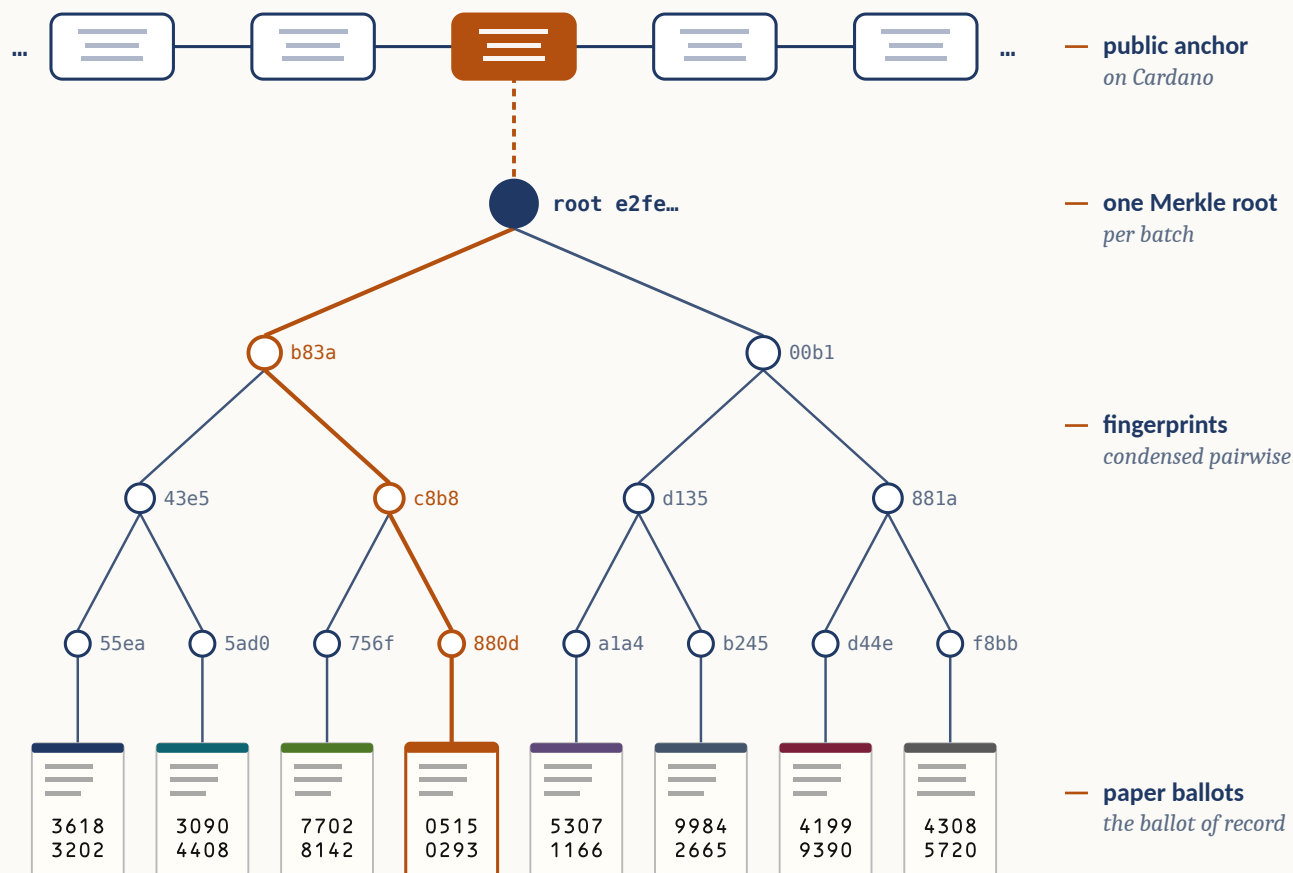


THE VOTE COIN PROTOCOL

A Multi-Layered, Verifiable Voting Framework



$$V_d = V_p$$

V_d the record the tablet keeps

V_p the paper the voter reads

each a salted SHA-256 fingerprint · Section VIII

*The paper proves the chain.
The chain indexes the paper.*

Voter-verified paper ballots remain the legal record. Their fingerprints are condensed into one Merkle root per batch and anchored on a public ledger, so the audit record can be checked by anyone and cannot be quietly rewritten later.

Prepared for Technical Review and an Early Consultation
with the U.S. Election Assistance Commission

Principal Architect: **Michael Twigg, PhD**

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How to Read This Document

This white paper is written for two audiences at once: technical evaluators (election security professionals, test laboratories, and cryptographers) and informed citizens who want to understand what is being proposed and why. To serve both, the document follows two rules throughout.

First, every technical term is defined in plain language the first time it appears, directly in the sentence where it is used. Acronyms are spelled out at first use. A complete Glossary is provided at the end for reference. Sources are cited by author and year, and standards, statutes and court cases by their designation; all appear in the References.

Second, the document is organized into eight color-coded Parts. Each Part opens with a divider page that lists who it is for, its sections and its figures. Every section carries a colored banner matching its Part, so the reader can navigate by topic at a glance:

Color Theme	Covers
Navy — Foundation	Part One: executive overview, the problem being solved, and the five verification layers (Sections I–III)
Teal — Architecture	Part Two: how the system is physically and digitally built (Section IV)
Burnt Orange — Security & Risk	Part Three: threat model, governance and oversight, disaster recovery, and reconciliation (Sections V–VIII)
Green — Privacy & Law	Part Four: privacy impact assessment and the legal and regulatory framework (Sections IX–X)
Purple — Operations & Scale	Part Five: comparative analysis, performance, throughput, voter onboarding, and implementation (Sections XI–XIV)
Slate — Certification & Rights	Part Six: integrity invariants, testing schedule, the Voter Bill of Rights, and the request for an EAC technology consultation (Sections XV–XVIII)
Burgundy — Objections & Rebuttals	Part Seven: fifty anticipated criticisms of this protocol, answered directly (Section XIX)
Gray — Appendices	Part Eight: about the architect, Glossary, and References

One question should stay in the reader’s mind from the first page to the last: *what, exactly, is the voter being asked to trust?* The honest answer for most current systems is “quite a lot, on faith.” The answer this protocol is engineered to give is “almost nothing” — a paper ballot the voter reads with their own eyes, and mathematics that anyone is free to check.

Status as of September 27, 2026 (Revision 4). Draft standards, court cases, the Election Assistance Commission’s membership and Cardano’s network settings (its protocol parameters) can change after publication. Each is stated as of September 2026 and cited to its source; anyone applying this paper should confirm the current status there first.

The Vote Coin Protocol is not affiliated with any cryptocurrency, including the unrelated “VoteCoin (VOT)” token.

FIGURE 1 · HOW TO READ THIS DOCUMENT

How to read the figures

One ballot, 0515 0293, is followed wherever a ballot appears

THE TRACED BALLOT



0515 0293 is traced in orange wherever it appears
the other seven keep the tab colors they have on the cover

WHAT EACH COLOR MARKS

- Ballot 0515 0293 only: its record, fingerprint and path
- Structure, identity, and whole-batch values like the root
- Labels, and “what if” values that did not happen
- Ab Anything printed on paper
- The band at the top lights the current Part

HOW NUMBERS APPEAR

In full, where a reader should check them:

**880d738d 9f978125 7ab513b3 12e5581f
71419543 b9c74f7e 5d6b7634 08e537d6**

64 hexadecimal digits in eight groups of eight

Shortened, inside step-by-step working:

880d738d...08e537d6

As printed on the ballot, in OCR-B:

0515 0293 the validation index

SYMBOLS



dashed line: no network
connection in or out



write-once media:
how data crosses



one of three keys,
each held separately



blocks on Cardano,
the public anchor



credential card:
identity, never linked



the voter's receipt:
a tracking code

The voter's identity is drawn in navy and never touches 0293's path: no log, time or sequence ties a ballot to a voter.
Each page ends as the cover does: a key (an equation or a count), a takeaway, and a “Check it” line.

$$V = H(R \parallel s)$$

R the ballot record: index and selections, as text

s a secret 32-byte salt, fresh for each ballot

$\parallel$ joined end to end · H SHA-256

V the fingerprint: 64 hexadecimal digits

*Follow one ballot.
Check every number.*

Every ballot value comes from one frozen sample batch of eight ballots — the batch on the cover. A real batch holds at least 100 ballots (fewer only at the smallest Nodes); eight fit on a page.

Check it: 0293's record R is 70 bytes; with its 32-byte salt the input is 102 bytes, and SHA-256 of it is 880d738d...08e537d6. Every input is listed in *Figures: Worked Examples*.

PART ONE

Foundation

Sections I – III · Executive Overview · Problem Statement · The Five Pillars

WHO THIS PART IS FOR

Every reader, regardless of background. No technical knowledge is assumed. Begin here — the rest of the document builds on these three sections.

IN THIS PART

I. Executive Overview..... 7

II. Problem Statement & Current Landscape..... 9

 2.1 Known Failure Modes in Current Infrastructure..... 9

 2.2 Why Blockchain Alone Is Not the Answer..... 9

III. The Five Pillars of Verification..... 10

FIGURES IN THIS PART

Figure 2 · The life of one ballot..... 8

Figure 3 · The five pillars of verification..... 12

I. Executive Overview

The United States asks its citizens to trust election infrastructure most of them have never seen, cannot inspect, and could not audit if they tried. Much of that infrastructure is proprietary, aging, and certified to standards written before modern threats existed. The result is predictable: every close election now ends in a dispute about the machinery itself. The Vote Coin Protocol is built on a different premise — that no one should have to *trust* a voting system, because everyone should be able to *verify* it.

The protocol uses the **Cardano blockchain** — a public, decentralized ledger; in plain terms, a permanent record book that no single company, agency, or person controls or can rewrite — as its audit backbone, or “Truth Layer.” Onto that ledger it introduces the **Vote Coin**: a single-use digital token whose only value is the right it represents — one citizen, one ballot, one election. It is not money. It cannot be traded, sold, or transferred, and it is burned once its ballot is cast.

Every digital vote is paired with a mandatory physical ballot, printed inside a **Secure Node** — a hardened polling facility that is **air-gapped** (with no network connection in or out) and **electromagnetically shielded (to a published, testable standard**, so that electronic signals cannot leak out of the room). The digital record and the paper ballot are cryptographically bound to each other in a one-to-one handshake, and each can be used to audit the other. The paper ballot — not the blockchain — remains the legal ballot of record.

The protocol is designed to satisfy the **Voluntary Voting System Guidelines 2.0 (VVSg 2.0)** — the federal benchmark for voting systems, adopted by the U.S. Election Assistance Commission (EAC) in February 2021 — and directly addresses the principles most demanding for any next-generation system: auditability, software independence, system integrity, accessibility, security, and transparency. It is also tracked against the draft VVSg 2.1 (the September 2025 revision, whose wording and requirement numbers may change before adoption), which bars encoded voter selections on paper records and requires a unique identifier on each one — both features of this design. The EAC has had no commissioners since July 9, 2026, so the draft cannot be adopted until new commissioners restore a quorum; staff testing and certification continue under the Program Manual (EAC, 2026c; CRS, 2026). It further aligns with **HAVA Section 301** (the Help America Vote Act’s core equipment requirements), **FIPS 140-3** (the Federal Information Processing Standard for cryptographic modules, successor to FIPS 140-2), **WCAG 2.1 AA and 2.2 AA** (the Web Content Accessibility Guidelines for digital interfaces), and the security controls of **NIST SP 800-53** (the National Institute of Standards and Technology’s control catalog).

Governance rests on a **Tripartite Oversight Triad**: three independent institutions, each holding one of three cryptographic authorization keys, with every privileged action requiring all three. In the reference model these are the state election authority (custody), an accredited civilian technical-certification body, and a bipartisan citizen oversight board, and Section VI shows how the same three-key pattern is instantiated with jurisdiction-appropriate institutions, since American elections are administered by states, not the federal government. No single actor — including the architect — can access or alter voting data, or release the audit record, alone; certifying results stays with the canvassing authority under state law.

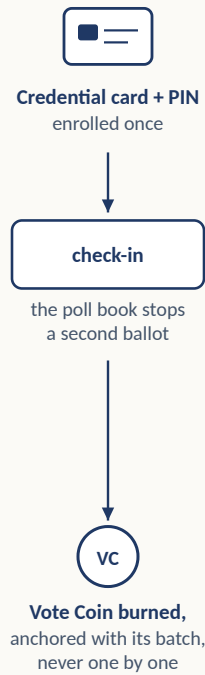
This design directly answers the failure modes that peer-reviewed election-security literature identifies in prior electronic and blockchain voting proposals: endpoint compromise, coercion and vote-selling, insider threat, and single points of failure. Section XIX confronts fifty of those criticisms individually.

FIGURE 2 · SECTION I · PART ONE

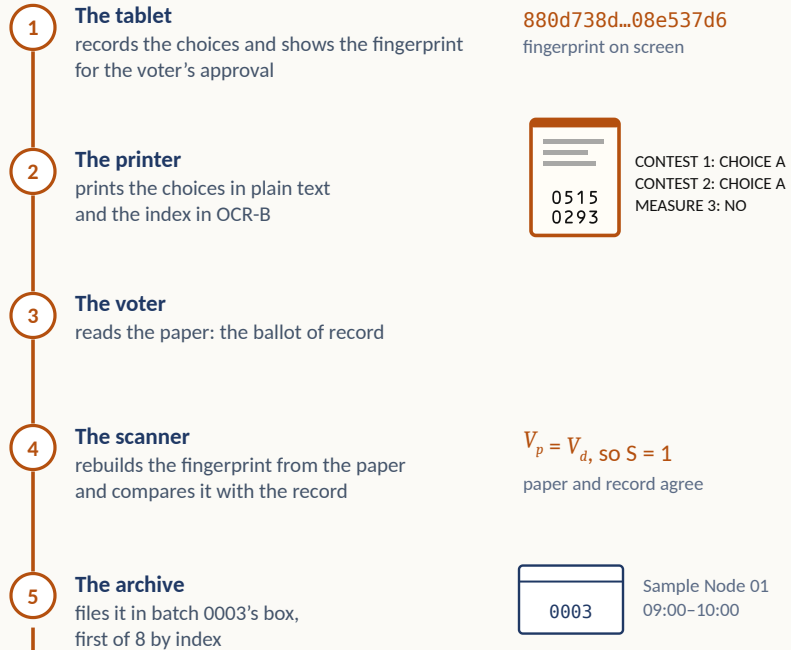
The life of one ballot

Ballot 0515 0293, from check-in to a public anchor anyone can check

WHO MAY VOTE



WHAT WAS VOTED — BALLOT 0515 0293



The batch

sorted fingerprints → one Merkle root, signed inside the Node with sequence number 0003

root e2fe3a46...f8d72a75

carried out on write-once media

*burns join the chain only as a batch,
in a crowd of 100+, or a small Node's total*

no link



$$V_d = V_p$$

V_d fingerprint of the tablet's record

V_p fingerprint of the paper, rebuilt by OCR

for 0293, both are 880d738d...08e537d6

and the receipt's tracking code is the same value

*Two records of one ballot,
checked against each other.*

Identity and ballot travel separate paths. The paper the voter read is the ballot of record; the fingerprint, the root and the anchor let anyone check that it was anchored.

Check it: from the receipt code 880d738d...08e537d6, the three hashes of Figure 16 rebuild the root e2fe3a46...f8d72a75 anchored for batch 0003.

II. Problem Statement & Current Landscape

American election administration rests on a patchwork of systems that vary widely in age, technology, and security posture. Nearly all equipment in use today was certified to standards adopted in 2005 and 2015 (VVSG 1.0 and 1.1) or earlier. The first systems certified to VVSG 2.0 arrived only in 2025–26; the EAC expects wide use around 2029 and estimates the cost of full replacement at \$2.71 billion (EAC, 2026a).

2.1 Known Failure Modes in Current Infrastructure

- **Proprietary opacity.** The source code and hardware designs of most deployed voting systems are trade secrets — though in September 2026 the EAC certified its first open-source system, showing that openness can pass federal testing (EAC, 2026b). The public is asked to trust what it is forbidden to examine.
- **Inconsistent audit trails.** Only five states require risk-limiting audits (RLAs) by statute — Colorado, Georgia, Nevada, Rhode Island and Virginia — while about ten more run pilots or permit them (NCSL, 2024; Verified Voting, n.d.). An RLA is a statistical check of paper ballots rigorous enough to confirm that the reported winner actually won.
- **Supply-chain risk.** Hardware manufactured or assembled overseas can be compromised before it ever reaches a warehouse, in ways no post-election audit can detect.
- **Single-vendor dependency.** A handful of vendors serve thousands of jurisdictions. One compromised or failed vendor becomes a national event.
- **Physical custody variability.** Chain-of-custody practices for ballots, memory cards, and tabulators differ from county to county, complicating any audit that crosses jurisdictional lines.

2.2 Why Blockchain Alone Is Not the Answer

A substantial body of peer-reviewed research — including work by the National Academies of Sciences (2018), by Park, Specter, Narula, and Rivest in the *Journal of Cybersecurity* (2021), and by Specter, Koppel, and Weitzner on a deployed blockchain voting app (2020) — concludes that blockchain technology, by itself, does not solve the security requirements of public elections. That research is correct, and this protocol is built in agreement with it, not in defiance of it:

- **Software dependence remains.** A blockchain records whatever it is given. If the software generating the vote record is compromised, the ledger will immutably preserve a fraudulent input.
- **The endpoint problem.** Voter-facing devices sit outside the blockchain and are the highest-risk attack surface. Compromise there produces fraud the chain cannot see.
- **Receipt-based coercion.** Any system that hands a voter proof of *how* they voted enables vote-buying and intimidation. Verifiability must never reveal ballot content.

Accordingly, the Cardano blockchain is **not** this protocol's security mechanism. It is the immutable audit backbone. Security comes from the combination of in-person voting, physical air-gapping, multi-factor identity verification, mandatory paper ballot production, three-party governance, and privacy-preserving cryptography — with the blockchain making every one of those layers publicly checkable.

⚠ Key Boundary: This protocol involves **no internet voting and no remote voting of any kind**. Every vote is cast in person at a Secure Node. This single design decision eliminates the entire class of remote-access attacks that peer-reviewed literature identifies as the fatal flaw of online voting.

III. The Five Pillars of Verification

Integrity here is achieved through redundancy. The protocol applies the security engineer's maxim — “don't trust; verify” — across five independent, sequential layers. A fraudulent outcome would have to defeat several independent layers at once — including the voter's own reading of the paper.

1. **Identity Layer (Credentialing).** Voters authenticate with **multi-factor authentication (MFA)** — proving identity through several independent means rather than one — built on **Hyperledger Identus** (formerly Atala PRISM), a **decentralized identity (DID)** framework in which the voter, not a central database, holds their own credentials. At the polls the voter presents a credential card issued at enrollment and unlocks it with a PIN. A voter who opted in may add a face match against the template stored on the card. Email is not used as a factor — NIST SP 800-63B-4 bars it — and no phone is required. Eligibility itself always comes from the state's voter registration database. A voter without the card, or whose card fails, is never turned away: the ordinary poll-book check-in and a paper ballot remain available, exactly as today — the credential card is a convenience and an audit aid, never a condition of the right to a paper ballot.
2. **Ledger Layer (Cardano).** Each ballot's salted fingerprint — a SHA-256 hash of the ballot record mixed with a secret random value, so the fingerprint identifies the record without revealing it (Section 8.1) — is recorded inside the Secure Node the moment the ballot is cast, then committed to Cardano in batches: each batch's fingerprints, up to 4,096, are condensed into one Merkle root — a single fingerprint of the whole batch that any one ballot can still be proved part of — and anchored in a single transaction under Cardano's **Extended Unspent Transaction Output (EUTXO)** model (Chakravarty et al., 2020) — an accounting design in which a transaction's outcome can be computed *before* it is submitted. The result is a public, time-ordered anchor for every ballot, with Cardano never in the voter's path (Section XII).
3. **Client Verification.** Before the digital handshake finalizes, the local client independently re-derives the ballot's fingerprint and displays it for the voter's approval. The voter sees and approves the fingerprint of their own record — the tracking code their receipt will carry.
4. **Physical Audit Trail.** An OCR-indexed paper ballot — printing the voter's selections in **human-readable text alongside a human-readable validation index (set in OCR-B, a typeface designed to be read by machines and people alike, and read by optical character recognition, not a barcode or QR code)** — is produced inside the Secure Node. The validation index is a random number generated inside the Node; no log, timestamp, or sequence ties it to the voter, the session, or the check-in record (Section 8.1; VVSG 2.0, Principle 10). The ballot satisfies HAVA's requirement for a permanent paper record with a manual audit capacity (52 U.S.C. § 21081(a)(2)) and the voter-verified paper requirements many states impose, Tennessee's among them (Tenn. Code Ann. § 2-9-101(d), in force since January 1, 2024). Once printed and checked by the voter, it can be counted without any computer.

5. **Sovereign Handshake.** A second, independent robotic scanner reads the deposited physical ballot — its printed selections and validation index — checks it against the digital record (Section VIII), and files it in a numbered location — closing the loop so that the paper proves the chain and the chain indexes the paper.

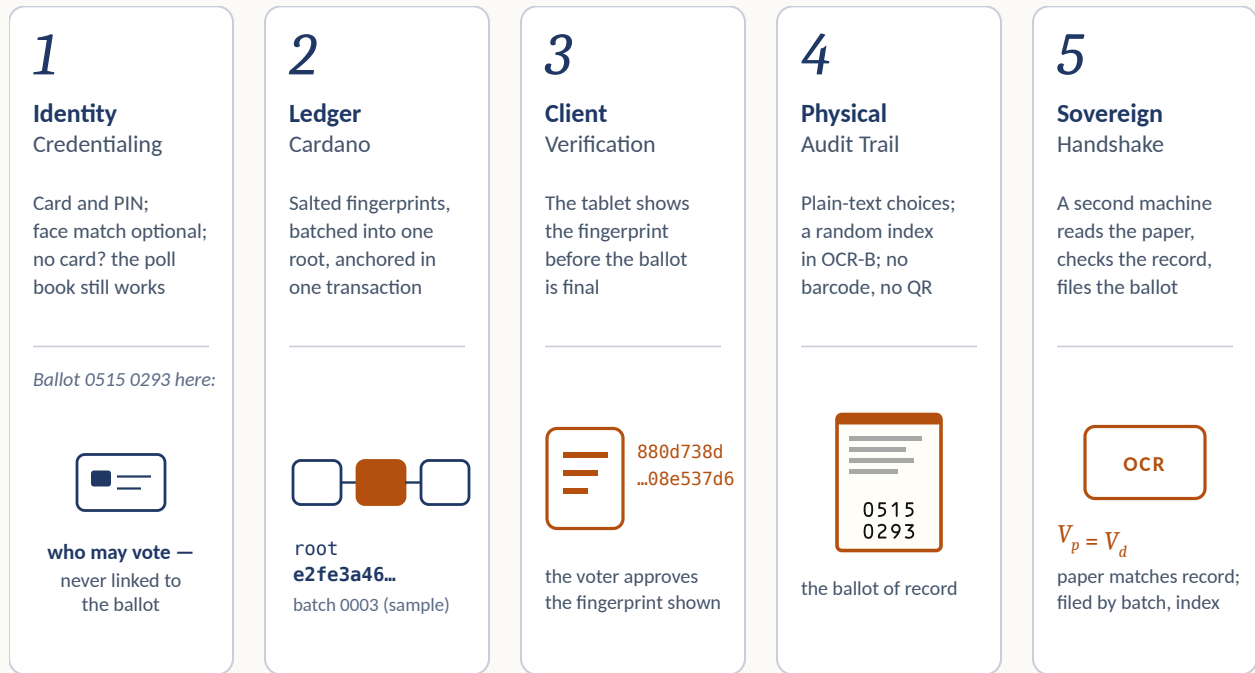
i One vote, start to finish: The five pillars meet a voter in this order. Each step is specified in the section named; Figure 2 draws the path of one ballot and Figure 6 draws what crosses the Secure Node's boundary.

1. **Enrollment (Section 13.1).** During the pre-election window the voter enrolls at a community Secure Node under the jurisdiction's existing registration and identification law and receives a credential card with a PIN; an on-card face match is optional. The card lasts across elections.
2. **Issuance (Section 14.1).** One Vote Coin is minted for each registered voter and held at a burn-only validator address. The voter never holds it, sees it, or pays for it.
3. **Check-in (Pillar 1; Section 4.3).** On Election Day the voter presents the card and PIN at a Secure Node; without a card, or with one that fails, the ordinary poll-book check-in applies. No one is turned away.
4. **Marking (Pillar 4; Section 8.1).** A tablet records the choices and prints the ballot — the selections in plain text beside a random eight-digit validation index, with no barcode or QR code. A pre-printed ballot marked by hand is always available.
5. **Review (Objection 7).** The voter reads the printed ballot. A wrong ballot is spoiled and reprinted.
6. **Deposit and reconciliation (Pillar 5; Section 8.2).** A second scanner reads the paper back, and the ARP compares its salted fingerprint with the tablet's record. A match files the ballot in its batch's archive box; a mismatch returns the ballot to the voter to mark again.
7. **Receipt (Section 15.2).** The voter leaves with a tracking code — the ballot's fingerprint — that proves inclusion and reveals nothing about the choice.
8. **Batching (Section 8.1).** Once an hour the Node sorts the batch's fingerprints, builds their Merkle root, signs the batch record, and writes it to write-once media, carried out under two-person custody.
9. **Anchoring (Sections 4.3 and 12.1).** The anchoring workstation, outside the certified voting system, publishes the root on Cardano and burns the batch's Vote Coins (from Phase III, it records their credential tags). Nothing about any vote leaves the Node.
10. **Counting (Section 12.1).** Unofficial totals and the official canvass come from the paper, as today. The chain counts nothing.
11. **Verification and audit (Sections 13.2, 14.1 and 15.2).** Anyone can check a receipt's code against an anchored batch on the public verification site; any accredited auditor can have any anchored batch's paper retrieved in seconds; and after certification the unused coins are burned in the close-out and the count published.

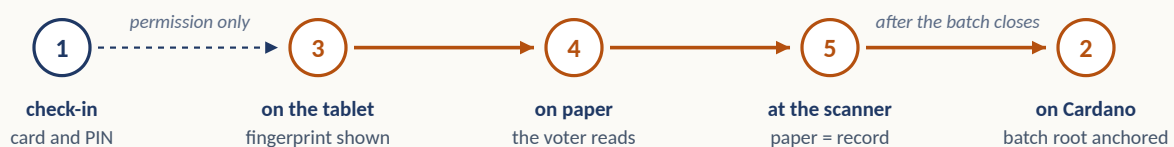
FIGURE 3 · SECTION III · PART ONE

The five pillars of verification

What each layer does, and what ballot 0515 0293 looks like there



IN THE ORDER THEY HAPPEN, FOR ONE BALLOT



To change an outcome, fraud would have to beat several layers at once — including the voter reading the paper.

5 independent layers

- 1 decides who may vote
- 2 makes the audit record public
- 3 shows the voter the recorded fingerprint
- 4 is the ballot of record
- 5 checks the paper against the record

*Don't trust. Verify —
five times over.*

Numbered as in Section III. For one ballot the layers run 1, 3, 4, 5, then 2: the anchor comes last, after its batch and the next one have closed. Layer 1 passes permission only — nothing about the voter travels with the ballot.

Check it: the fingerprint on the tablet (layer 3) and the one rebuilt from paper (layer 5) are both 880d738d...08e537d6 — recomputed in Figure 11.

PART TWO

Architecture

*Section IV · System Architecture & Operations***WHO THIS PART IS FOR**

Technical evaluators, systems and security engineers, election-equipment vendors, and Cardano developers assessing feasibility.

IN THIS PART

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4.3 The Air-Gapped Secure Node & Redundant Hardware.....	16
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IV. System Architecture & Operations

4.1 The Economic Model: A Zero-Cost Utility Token

- **Fees are public money.** Anchoring fees and token deposits are paid from the jurisdiction's appropriated election budget — the cost is small (Objection 21) — and held in a multi-signature account controlled by public officials. The voter pays nothing, ever; there is no fee, no barrier, and nothing that could be construed as a poll tax. No private or foundation money funds election operations.
- **Account control.** The election account is a multi-signature address — a wallet that mathematically refuses to move funds unless several independent keyholders sign together — held by the jurisdiction's public officials under state public-funds law. No unilateral withdrawal is possible, and no private party holds a key.

4.2 The Vote Coin Distinction: A Non-Monetary Asset

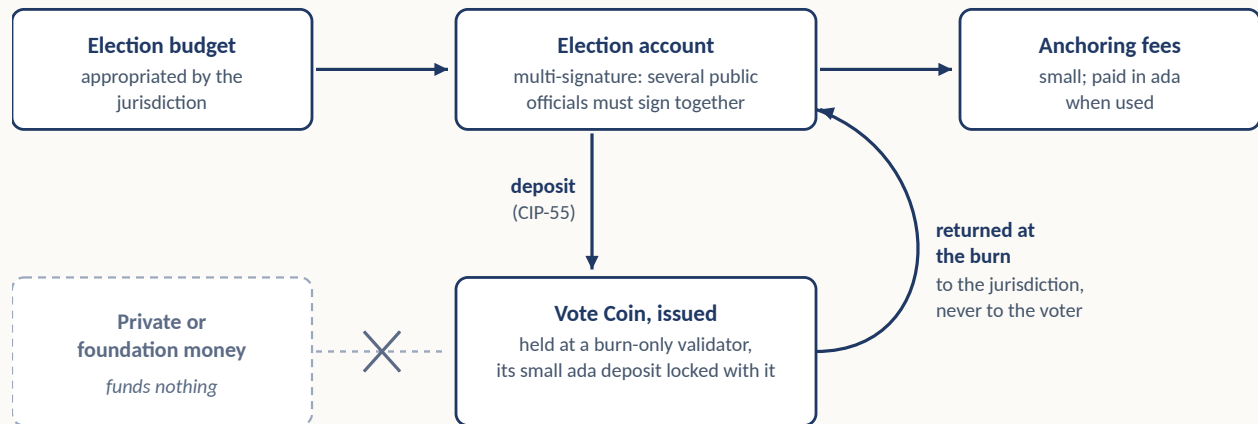
- **Non-monetary by construction.** Vote Coin's only "value" is the right to cast a single ballot. It carries no exchange rate, no denomination, and no market — because the rules that create it make a market impossible.
- **Non-transferable by script.** Each Vote Coin is issued into an output locked by a **Plutus validator** — an on-chain rule that permits exactly one action: burning the token, on a Secure Node's signed record that the ballot was cast — or, once the election is certified and its challenge period has closed, in the three-key close-out burn of every coin left unused (Section 14.1). The minting policy governs creation and destruction; the validator governs custody. No wallet, including the voter's, can send it anywhere, so a Vote Coin cannot be sold, lent or pooled.
- **Finality by burning.** When the ballot is cast, the token is **burned** — permanently destroyed on-chain — leaving an indelible public proof that this voting right was exercised exactly once. Check-in — the precinct list or the jurisdiction's electronic poll book — still stops a second ballot at the moment of voting; the burn makes that fact publicly provable afterward. Burning a Vote Coin has no effect on the supply of ada (Cardano's currency) or on any monetary system.
- **Deposits stay public money.** Cardano requires every output holding a token to carry a small ada deposit (Cardano Improvement Proposal 55, CIP-55). As of September 2026, the deposit is 0.00431 ada for each byte the output takes up, counting a fixed 160 bytes of overhead — about 1 to 1.5 ada for one Vote Coin, depending on how its output is encoded. That rate is a **protocol parameter** — a network-wide setting that Cardano's on-chain governance can change — so check its current value when applying this design (Cardano mainnet protocol parameters, 2026). The jurisdiction funds those deposits, they stay locked with the token, and the burn returns them to the jurisdiction's account — never to the voter. No voter ever receives anything of value for voting (52 U.S.C. § 10307(c)).

FIGURE 4 · SECTIONS 4.1–4.2 · PART TWO

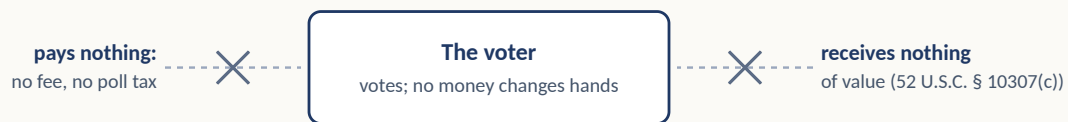
Public money in, nothing of value out

How fees and deposits are paid, and why a Vote Coin cannot become money

WHERE THE MONEY GOES



THE VOTER



WHY A VOTE COIN IS NOT MONEY

- 1 **No market**
no exchange rate, no denomination, no price
- 2 **No transfer**
no wallet, the voter's included, can send it
- 3 **One action**
burn, on a Node's signed record that a ballot was cast
- 4 **No effect on ada**
burning it changes no supply of any currency

Identity and money stay navy here: a Vote Coin is never linked to any ballot, including 0515 0293.

0 paid by or to any voter

fees from the appropriated budget
deposits go out at issue, back at the burn
account multi-signature, public officials
token one permitted action: burn

*Public money in; deposits come back.
 Nothing of value reaches a voter.*

Cardano requires a small ada deposit with every output that holds a token (CIP-55). The jurisdiction funds those deposits, and each burn returns its deposit to the jurisdiction's account.

Check it: on the public ledger, every burn transaction returns its deposit to the election account's multi-signature address — never to a voter.

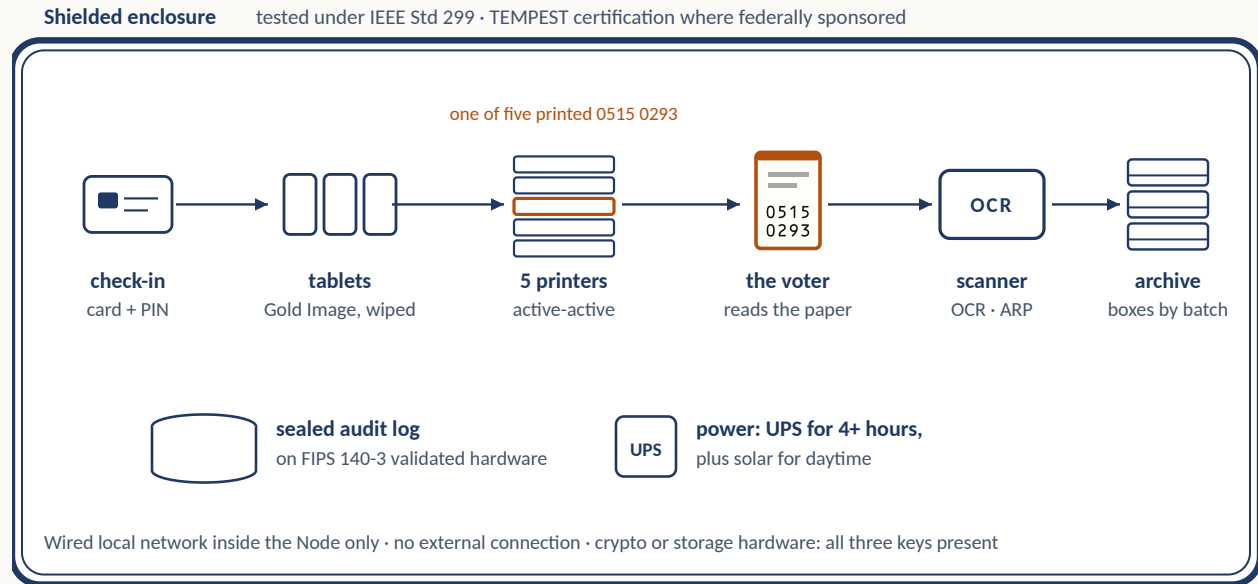
4.3 The Air-Gapped Secure Node & Redundant Hardware

- **Secure Node architecture.** A physically hardened facility, electromagnetically shielded to a published, testable standard (shielding effectiveness measured under IEEE Std 299, with TEMPEST certification — the U.S. government’s emission-security program — where a jurisdiction has federal sponsorship), with no network connection to anything outside it — a **full air gap**, as VVSG 2.0 requires (Requirement 14.2-E). Remote code execution and command-and-control attacks need a network path; here, none exists. Bulk data crosses the boundary only on write-once media, in two controlled steps: the election definition comes in before polls open (Objection 4), and signed batch records go out for anchoring (below). Two further crossings happen once per voter and never leave the Node’s physical perimeter: the credential card is read in at check-in, and a paper receipt is printed out. Neither is a network path. A networked electronic poll book, where a jurisdiction uses one, stays outside the Secure Node: what reaches the Node is the poll worker’s check-in decision, not data, and nothing about the ballot goes back.
- **Anchoring path.** The Secure Node never runs a networked Cardano node. It signs each batch record — the batch’s root and sequence number, its sorted fingerprints and its token burns — with Ed25519, the signature scheme Cardano itself uses (IOG Cryptography Handbook, n.d.), with migration to FIPS 204 (ML-DSA) on NIST’s post-quantum timeline (Objection 20), and writes it to fresh write-once media, which staff carry under two-person custody to an **anchoring workstation** outside the certified voting system. The workstation wraps the signed root, and the batch’s token burns, in Cardano transactions and submits them over two independent connections from different providers. Because the root is signed inside the Node, a compromised workstation can delay an anchor but cannot alter one, and a withheld anchor shows as a gap in the sequence. Anyone can check each published root against the Node’s signature. (If the EAC determines, through its Request for Interpretation process — the mechanism by which a manufacturer or test laboratory obtains a definitive reading of a VVSG requirement (Program Manual v3.0, §9) — that a hardware one-way data diode falls within the voting-system boundary Requirement 14.2-E permits, a diode can replace the carried media.)
- **Stateless voter tablets.** Each voter-facing tablet boots from a cryptographically signed, read-only **Gold Image** — a master copy of the operating system that cannot be altered — and is wiped and reloaded after every session. Nothing persists between voters: no cache, no credentials, no contamination.
- **Printer array (five printers, active-active).** Five archival-grade industrial printers operate in an active-active configuration: all five share the queue under load, and the failure of any single unit reduces capacity by one-fifth without interrupting a single voter. Any three can fail and the remaining two (600 ballots/hour) still exceed the Phase I peak of 500. Ballot stock is archival-rated to remain legible well beyond the federal 22-month retention requirement.
- **Tripartite maintenance.** Any intervention that touches a cryptographic module, data storage, the ballot archive or the air-gap boundary inside the Secure Node requires the physical co-presence of representatives of all three Triad institutions. Routine mechanical maintenance — clearing a printer jam, replacing paper or toner — is done by the Node’s systems technician in view of two poll workers of different parties and logged, so that voting is never held for a mechanical fault. No technician, vendor, or official touches the cryptographic or storage hardware alone. Technical certification — shielding and air-gap checks — is completed before voting opens, and no uniformed or armed personnel are present while polls are open.

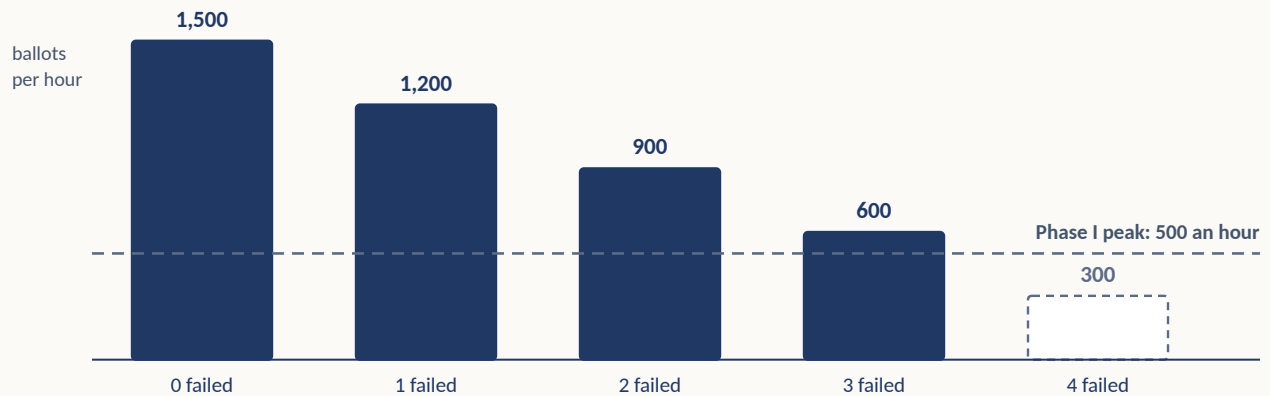
FIGURE 5 · SECTION 4.3 · PART TWO

Inside a Secure Node

The equipment, the power, and what happens when printers fail



PRINTING CAPACITY WITH FAILED PRINTERS



$$C(f) = (5 - f) \times 300$$

f printers failed, out of 5

C ballots printed per hour; one printer makes about 300

$C(3) = 600$, still above the Phase I peak of 500

$C(4) = 300$, below it: lines would grow

*Any three printers can fail,
and printing still beats the peak.*

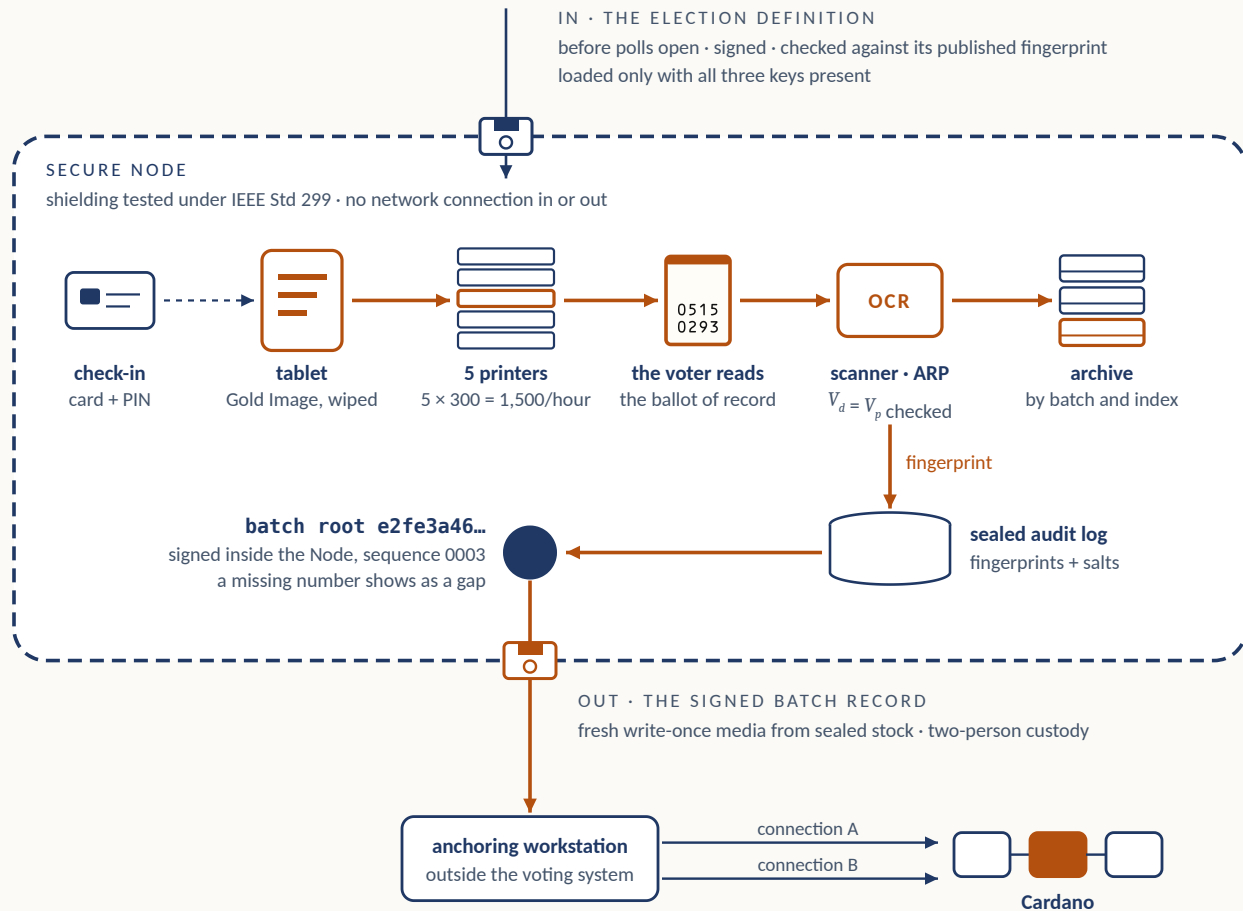
All five printers share the queue under load, so one failure costs a fifth of capacity, never downtime. The tablets boot from a signed, read-only image and are wiped after every voter.

Check it: $(5 - 3) \times 300 = 600 \geq 500$; $(5 - 4) \times 300 = 300 < 500$. With four or five printers working, filing is the slowest step, at 1,000 an hour (Figure 18).

FIGURE 6 · SECTION 4.3 · PART TWO

The air gap, and where data crosses

How data enters and leaves a Secure Node that has no network connection



Two providers, so no single company's network is a dependency. A compromised workstation can delay an anchor, not alter one: the root is signed inside the Node, and a withheld anchor shows as a gap in the sequence.

If an EAC Request for Interpretation finds a one-way data diode within 14.2-E, a diode can replace the carried media.

0 network connections

in or out of the Node, ever

2 bulk-data crossings, on write-once media

3 keys present to load the election

2 providers for the outside workstation

No network, in or out.

Bulk data crosses only on media.

VVSG 2.0 Requirement 14.2-E: a voting system must not be configured to “establish a connection to an external network” or to “connect to any device external to the voting system.” Section 18.3 asks the EAC to confirm the boundary.

Check it: the root carried out for 0293's batch is e2fe3a46...f8d72a75 — the root that Figure 16 rebuilds from 0293's fingerprint with three hashes.

4.4 Robotic Filing & the Physical Ledger

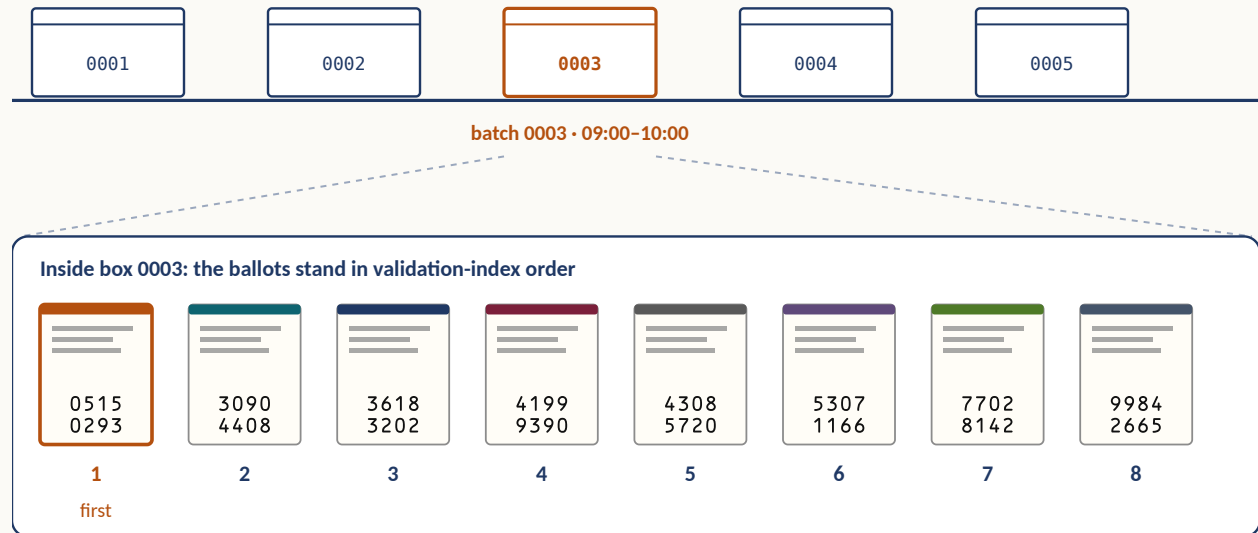
- **Indexed filing.** High-speed robotics scan each deposited ballot and file it in its batch's archive box, in validation-index order — so a batch's ballots are retrieved together, and any ballot is found by its batch and index. Given any anchored batch, the matching paper ballots can be retrieved in seconds; given any ballot, an auditor holding its sealed salt can locate its anchored fingerprint instantly.
- **An independent, non-digital record.** The filing archive is a complete election record in its own right. If every computer involved failed — including the Cardano network itself — the physical ledger remains intact, humanly countable, and legally sufficient for recounts and litigation.
- **Shrinking the oracle problem.** Blockchains famously cannot verify facts about the physical world without trusting an outside reporter (the **oracle problem**). By binding each paper ballot's salted fingerprint to its anchored twin, the protocol replaces “trust the reporter” with “check the math” at the exact point where the digital and physical worlds meet.

FIGURE 7 · SECTION 4.4 · PART TWO

Filed by batch and index

Where ballot 0515 0293 goes, and how an auditor finds it again

THE ARCHIVE AT SAMPLE NODE 01



TWO WAYS TO FIND A BALLOT

From a ballot to the record:

box 0003, index 0515 0293 + its sealed salt → its fingerprint in the published list

From the record to the ballot:

an anchored batch → box 0003, all 8 ballots together, in seconds

TWO ORDERS, NEITHER OF THEM TIME

In the box, by index: 0515 0293 is 1 of 8
In the tree, by fingerprint: 0515 0293 is 4 of 8 (Figure 12)

Indices are random and fingerprints are salted,
 so neither order says who voted when.

A repeated index is discarded and drawn again; used indices are kept as an unordered set, with no order or time.

$$P \approx 1 - e^{-n(n-1)/2N}$$

P chance some index repeats among n draws
 $N = 100,000,000$ possible indices
 $n = 1,000$: **0.5%** · 5,000: **11.7%** · 10,000: **39.3%**
 so the rule redraws; it never reuses

*Two numbers find any ballot:
 its batch and its index.*

The robot files each ballot in its batch's box, in index order. Given an anchored batch, its paper ballots come out together; given a ballot and its sealed salt, its fingerprint is found at once.

Check it: sorted by index, 0515... comes first of 8; sorted by fingerprint, 880d... is 4 of 8, after 55ea..., 5ad0... and 756f....

4.5 Accessibility & Universal Design

- **Accessibility standards.** Voter-facing interfaces meet WCAG 2.1 Level AA — the standard the Justice Department’s ADA Title II rule sets for state and local governments (DOJ, 2026) — and target WCAG 2.2 AA, the current W3C Recommendation, also approved as ISO/IEC 40500:2025 (W3C, 2025), exceeding the WCAG 2.0 AA baseline in VVSG 2.0: sufficient contrast, screen-reader compatibility, keyboard navigability, and scalable text.
- **Haptic and audio guidance.** Voice-guided flows and vibration-confirmed selections serve voters with visual or motor impairments, consistent with the Americans with Disabilities Act (ADA) and Section 508 of the Rehabilitation Act.
- **Language access.** Multi-language ballot rendering complies with Section 203 of the Voting Rights Act, which requires language assistance where minority-language populations exceed statutory thresholds; language coverage is a configuration setting, updated whenever the Census Bureau publishes new Section 203 determinations (U.S. Census Bureau, 2021).
- **Cognitive accessibility.** Simplified flows and plain-language instructions support voters with cognitive disabilities, consistent with VVSG 2.0 Principles 5 and 6: all voters can use the system regardless of ability, and can mark, verify, and cast their ballot privately and independently.

PART THREE

Security, Governance & Resilience

Sections V – VIII · Threat Model · Tripartite Governance · Disaster Recovery · Automated Reconciliation

WHO THIS PART IS FOR

Security professionals, red teams, election officials responsible for custody and continuity, and anyone evaluating what happens when things go wrong.

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V. Threat Model & Risk Analysis

The threat model below follows the **STRIDE** framework (Microsoft, n.d.) — an industry-standard taxonomy that examines a system against six attack categories: Spoofing, Tampering, Repudiation, Information disclosure, Denial of service, and Elevation of privilege — consistent with NIST SP 800-30 risk-assessment guidance and VVSG 2.0 security requirements.

5.1 Threat Table

Threat Category	Attack Vector	Protocol Mitigation	Residual Risk
Spoofing / identity fraud	Impersonating a registered voter at enrollment or at the ballot stage	Credential card unlocked by a PIN (something the voter has and knows), with an optional on-card face match; the ordinary poll-book check remains available	Low. Requires physical presence plus theft of the voter's card and knowledge of the PIN
Tampering / vote manipulation	Modifying vote data in transit or at rest	Full air gap (no network connection); signed, publicly anchored batch records; the ARP (Automated Reconciliation Protocol, Section VIII) check of the scanned paper against the digital record	Very low. The air gap removes remote tampering; the paper record catches anything the chain cannot
Repudiation	A voter or official denies that a vote was cast or counted	Anchored receipt (a Merkle inclusion proof — proof that the ballot's fingerprint is in an anchored batch); court-admissible physical ballot; ARP confirmation	Very low. Three independent records (DID log, anchored batch records, physical ballot)
Information disclosure / privacy	Linking a voter's identity to their ballot choice	Zero-knowledge proof confirms eligibility without revealing identity; ballots are anonymized before scanning	Low. Unlinkability is cryptographic, with physical anonymization as a second layer
Denial of service	Disabling a Secure Node, or the anchoring workstation, during voting	Five-printer redundancy; UPS and solar power; encrypted local storage; anchoring from any authorized workstation over two independent connections	Low. Voting never depends on a network; if anchoring is disrupted, signed records wait and are anchored later
Elevation of privilege / insider threat	A Triad member, vendor, or official acting unilaterally	Three-Key Rule: simultaneous physical presence of all three Triad representatives for any privileged access	Low. Requires collusion across three independent institutions

Threat Category	Attack Vector	Protocol Mitigation	Residual Risk
Supply-chain attack	Hardware or software compromised before deployment	Read-only Gold Image; tripartite acceptance inspection; shielded enclosures from vetted suppliers, with a published hardware bill of materials	Medium. Third-party hardware attestation is specified for Phase I to reduce this further
Coercion / vote-selling	A voter pressured or paid, or a receipt used as proof of choice	Receipts never reveal ballot content; the token is non-transferable; voting is supervised and in person	Low–medium. In-person supervision limits, but cannot fully eliminate, pressure applied outside the facility

5.2 Mitigations for Residual Risks

- **Supply chain.** Engage an independent, accredited hardware-security laboratory to certify all Secure Node components before Phase I — including a firmware audit of every printer and a print test confirming that no printer adds tracking marks (the near-invisible dot patterns many printers embed, which encode the printer's serial number and the date and time of the print; Richter et al., 2018) or any other covert marking that could tie a ballot to its moment of printing — and publish the hardware bill of materials for Triad review.
- **Coercion zone.** No recording is permitted inside the voting room, and the buffer around each Secure Node follows the host state's polling-place law (buffer rules vary by state).
- **Independent red team.** Federal law now requires the EAC to include penetration testing in certification testing (52 U.S.C. § 20971(e), added December 18, 2025). This paper asks for a consultation and a supervised pilot, not certification, so that duty does not yet reach the design; meeting its bar early is the point of commissioning an independent election-security red team to test the Secure Node and the anchoring workstation before any Phase III deployment, and publishing the results.

FIGURE 8 · SECTION V · PART THREE

Eight threats, eight answers

The threat table at a glance: what each attack faces, and what risk remains

THREAT (STRIDE, PLUS TWO)	MAIN ANSWER IN SECTION 5.1	RESIDUAL RISK			
		Very low	Low	Low-medium	Medium
S Spoofing impersonating a voter	card unlocked by a PIN; poll-book check stays				
T Tampering changing vote data	air gap; signed anchors; ARP checks paper				
R Repudiation denying a vote was cast	three records: DID log, anchor, paper				
I Information disclosure linking voter to ballot	zero-knowledge proof; anonymized ballots				
D Denial of service disabling a Node	five printers; UPS and solar; no network				
E Elevation of privilege an insider acting alone	three keys, three institutions				
+ Supply chain compromised pre-deployment	Gold Image; joint acceptance; published BOM				
+ Coercion, vote-selling pressure, or a receipt as proof	receipt reveals nothing; in person				

Supply chain carries the highest rating: third-party hardware attestation is specified for Phase I to lower it.
Beneath the tampering threats sits the paper ballot, which no remote attack can reach.

6 + 2 threats

STRIDE — the six STRIDE categories
+ supply chain, and coercion or vote-selling
ratings, as in Section 5.1: **2** very low, **4** low,
1 low-medium, **1** medium

*No threat is rated zero.
Every one has an answer.*

STRIDE is Microsoft's threat taxonomy (Microsoft, n.d.), used consistently with NIST SP 800-30. The ratings are the paper's own, from the residual-risk column of the threat table.

Check it: count the dots: 2 + 4 + 1 + 1 = 8 threats, matching the eight rows of the table in Section 5.1.

VI. Governance: The Tripartite Oversight Triad

Every privileged action in the Secure Node ecosystem — opening the ballot archive, servicing cryptographic or storage hardware, releasing the audit record — is governed by the Three-Key Rule: the simultaneous physical presence and cryptographic authorization of three representatives from three independent institutions. None can act alone. The Triad certifies the audit record, not the election: certifying results remains the canvassing authority’s legal duty, and no key-holder can delay it. If a key-holder fails to act by a set deadline, a statutory alternate holds that key.

6.1 The Reference Model

Entity	Core Responsibility	Key Function	Independent Authority
Custodial Authority	The state election authority: physical hardware integrity, site security, and chain of custody for Secure Node enclosures	Holds Key K(cust)	Never acts alone; a statutory alternate holds the key if it fails to act by the deadline
Technical Certification Body	An accredited civilian body: shielding and air-gap verification, completed before voting opens and away from the polling place	Holds Key K(tech)	Never acts alone; a statutory alternate holds the key if it fails to act by the deadline
Citizen Oversight Board	A bipartisan public-interest body: civilian oversight of robotic filing, audit access, and transparent results publication	Holds Key K(pub)	Never acts alone; a statutory alternate holds the key if it fails to act by the deadline

6.2 Jurisdiction-Appropriate Instantiation

The Constitution assigns election administration to the states, and federal law (including the Posse Comitatus Act, which restricts military involvement in domestic civil affairs) constrains which institutions may hold which roles, and 18 U.S.C. § 592 makes it a crime to bring troops or armed men to any place where an election is held. The Triad is therefore a *governance pattern*, not a fixed cast of agencies. What is non-negotiable is the structure: three keys, three genuinely independent institutions, no unilateral action. Who holds them adapts to law:

Deployment Level	Example Instantiation	Legal Basis
Municipal / county pilot	County sheriff’s office · State National Guard cyber unit (Title 32, state-controlled status) · League of Women Voters or equivalent civic body	State election code; Title 32 state activation
State deployment	State election authority · State Guard technical unit · Citizen oversight board with statutory standing	State constitution and election statutes

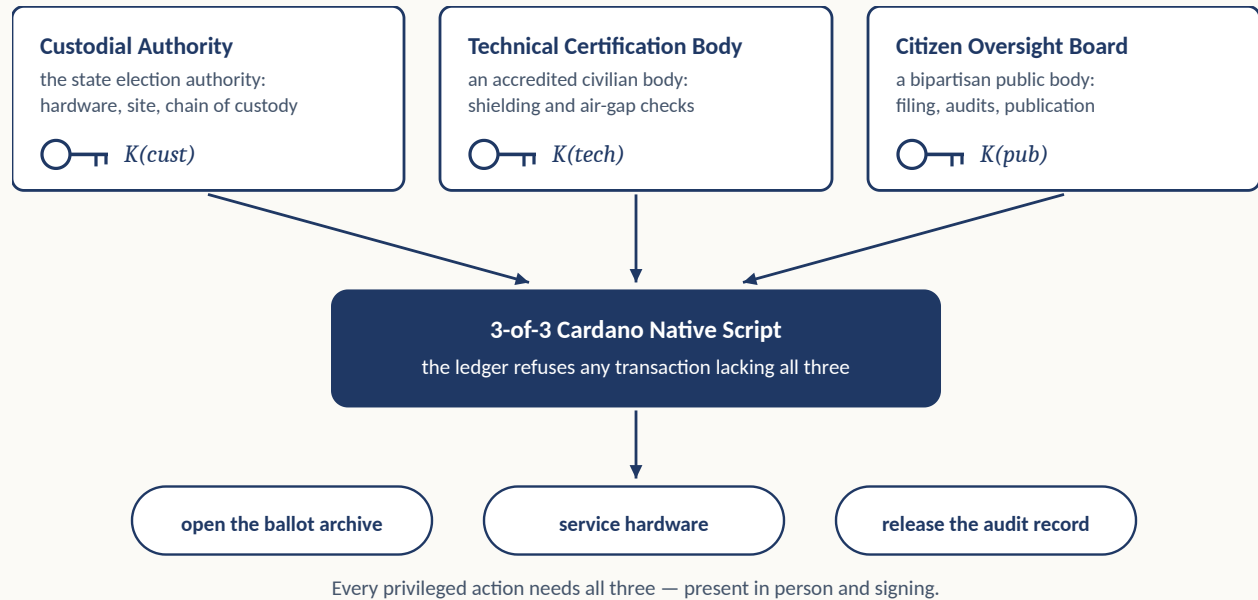
Deployment Level	Example Instantiation	Legal Basis
Reference model	State election authority (custody) · Accredited civilian technical-certification body · Bipartisan citizen oversight board	State election code. Federal agencies, where law permits, certify equipment before voting — never at the polling place

i Enforced by cryptography, not policy: The Three-Key Rule is implemented as a 3-of-3 **multi-signature** policy in a Cardano Native Script — meaning the ledger itself refuses any privileged transaction lacking all three signatures. There is no administrative override, no superuser, and no software bypass. On the chain, each seat can be written to accept its statutory alternate's key once the deadline passes — a time-locked condition Cardano's native scripts express with their after and before slot bounds, which the ledger itself evaluates when a transaction is submitted (IntersectMBO, n.d.). Off the chain — opening the sealed record or the archive — no clock is consulted: the statutory alternate simply attends in the key-holder's place, under the statute that names them. A native script can never be edited, and moving the election's authority or funds to a new script is itself a transaction that needs all three signatures — a self-reinforcing loop. This is by design.

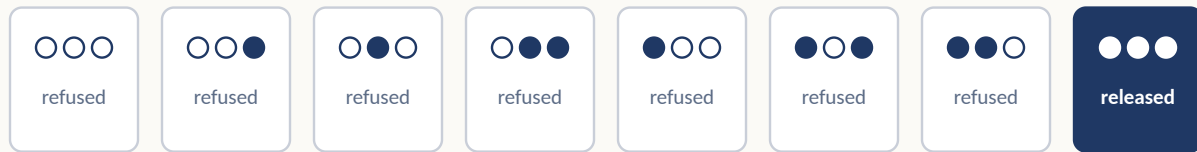
FIGURE 9 · SECTION VI · PART THREE

Three keys, three institutions

The Three-Key Rule: three key-holders present, three signatures on the ledger



ALL EIGHT WAYS THREE KEY-HOLDERS CAN SHOW UP



Dots: cust, tech, pub; filled = present and signing.

If a key-holder fails to act by a set deadline,

a statutory alternate holds that key — on chain, a time lock native scripts express with after/before slot bounds.

Outside the Triad: certifying results.

That stays the canvassing authority's legal duty, and no key-holder can delay it.

3 of 3 signatures

$$C \Leftrightarrow \text{Sig}(K(cust)) \wedge \text{Sig}(K(tech)) \wedge \text{Sig}(K(pub))$$

C a privileged action is released
 $\text{Sig}(K)$ a valid signature with that key; $\wedge$ and
 1 of the 8 combinations releases

Three keys, three institutions. None can act alone.

The reference model's three civilian bodies; Section 6.2 shows how a jurisdiction fills the three seats under its own law. A native script can never be edited; moving its authority to a new one needs all three keys.

Check it: three key-holders, each present or not, give $2 \times 2 \times 2 = 8$ combinations; only one has all three.

VII. Disaster Recovery & Continuity of Operations (COOP)

- **Local data persistence.** Secure Nodes always store vote packages — the digital record of each ballot, the same record R the tablet produced (Section 8.1), kept for reconciliation and audit — encrypted with AES-256 in GCM mode (the NIST authenticated-encryption mode, SP 800-38D) on FIPS 140-3 validated hardware. If the anchoring workstation or the Cardano network is unavailable, signed batch records wait and are anchored, timestamps preserved, when it returns. Voting continues uninterrupted throughout.
- **Power redundancy.** Every Node carries an uninterruptible power supply (UPS) rated for a minimum of four hours, plus a local solar array for indefinite daytime operation independent of the grid.
- **Hardware failover.** The five-printer array absorbs individual unit failures without interruption; critical components are dual-sourced and stocked on site. If the ballot scanner or the filing robot fails, ballots go into a sealed ballot box, are filed by hand in numbered sleeves, and are reconciled under the after-filing rule (Section 8.2) when the equipment returns — the paper is the record, so voting never stops (Objection 41).
- **Anchoring redundancy.** Signed batch records can be anchored from any authorized anchoring workstation, and each workstation uses at least two independent connections from different providers, so no single company's network is a dependency.
- **Degraded mode, never suspension.** If two or more redundancy systems fail at once, the Node switches to pre-printed, hand-marked paper ballots deposited into the same sealed archive. Staff alert the Triad over an encrypted channel outside the Node, and reconciliation catches up after restoration. Polls stay open.
- **Recovery testing.** Full disaster-recovery exercises are conducted annually and before every election cycle, with results reported to the Triad and documented publicly.

FIGURE 10 · SECTION VII · PART THREE

Degraded, never suspended

How a Secure Node keeps the polls open through each kind of failure

WHAT FAILS	WHAT THE NODE DOES	VOTING
 Normal day	five printers, full capacity 1,500 ballots an hour	● polls open
 One printer fails	the other four share the queue 1,200 an hour, one-fifth less	● polls open
 Up to three fail	at least two printers still run 600+ an hour, above the Phase I peak	● polls open
 The grid fails	UPS for at least four hours, solar for daytime	● polls open
 Cardano or the workstation is down	signed batches wait; timestamps kept; anchored when it returns	● polls open
 Two or more redundancy systems fail	hand-marked paper ballots go into the same sealed archive	● polls open

In degraded mode, staff alert the Triad over an encrypted channel outside the Node — the Node itself never connects — and reconciliation catches up after restoration. Disaster-recovery drills run every year and before every election.

0 hours suspended

printers: **any 3 of 5** can fail
power: UPS for **4+ hours**, plus solar by day
anchoring: **any** authorized workstation,
over **2** independent connections

Degraded, never suspended. The last fallback is paper.

Each failure has its own fallback, and none of them needs the Node to connect. The hand-marked ballots of degraded mode go into the same archive and are counted and audited like any other.

Check it: one failed printer leaves $(5 - 1) \times 300 = 1,200$ an hour, one-fifth less than 1,500; three leave 600, still above 500.

VIII. The Automated Reconciliation Protocol (ARP)

The ARP is the system’s self-auditing loop — a deterministic error check that runs automatically after every single ballot, confirming that the digital record and the physical ballot agree before any vote is considered final. Where a conventional audit samples a fraction of ballots weeks later, the ARP reconciles every tablet-printed ballot in real time.

8.1 ARP Logic

The comparison uses **SHA-256**, a NIST-approved hash function (FIPS 180-4) computed in FIPS 140-3 validated modules — a mathematical fingerprinting algorithm for which producing two different inputs with the same fingerprint is computationally infeasible. Cardano uses its own hash, Blake2b-256 (Aumasson et al., 2013), internally; the ballot check never depends on it. Let R be the canonical ballot record — the validation index and the voter’s selections, written in a fixed, published format — and s a fresh secret random value (a **salt**) generated inside the Secure Node for each ballot and kept only in its sealed audit log:

$$\begin{aligned} V_d &= \text{SHA-256}(R \text{ as recorded by the voter tablet} \parallel s) \\ V_p &= \text{SHA-256}(R \text{ as rebuilt by OCR from the printed ballot's human-} \\ &\quad \text{readable selections and validation index} \parallel s) \\ \\ S &= 1 \quad \text{if } V_d = V_p \quad \rightarrow \text{paper and digital record agree} \\ S &= 0 \quad \text{if } V_d \neq V_p \quad \rightarrow \text{discrepancy; the paper governs (§8.2)} \end{aligned}$$

Published formats. Every input to this check is fixed in advance, so anyone holding the same data can reproduce each fingerprint and each batch root:

- **Ballot record R .** UTF-8 text, one field per line, each line ending in a line feed. The first line is INDEX, a colon, a space and the eight digits of the validation index. Each following line is one contest, in ballot order: its name, a colon, a space and the selection (NO SELECTION if the contest was left blank), spelled exactly as in the election definition.
- **Validation index.** Eight digits drawn at random inside the Secure Node for each ballot (100 million possible values) and never repeated among that Node’s ballots in the election. A draw that repeats an index already used at that Node is discarded and drawn again; used indices are kept as an unordered set, with no order or time. It is not derived from the voter, the session, the check-in record or the time, so it cannot tie a ballot to a voter. It is printed in OCR-B as two groups of four digits, for example 0515 0293.
- **Salt s .** 32 random bytes (256 bits) from a NIST-approved random bit generator (SP 800-90A Rev. 1; SP 800-90C) in the Node’s FIPS 140-3 validated module, made fresh for each ballot and kept only in the sealed audit log.
- **Batch tree.** The Merkle tree defined in IETF RFC 9162, Section 2.1.1, computed with SHA-256: each leaf is $\text{SHA-256}(0x00 \parallel \text{fingerprint})$ and each inner node is $\text{SHA-256}(0x01 \parallel \text{left} \parallel \text{right})$, so an inner node cannot be passed off as a ballot. The fingerprints are sorted in ascending order before the tree is built, so a batch reveals nothing about the order in which ballots were cast.

- **Batches.** A batch closes after 60 minutes or 4,096 ballots, whichever comes first — but never with fewer than 100 ballots: a short batch stays open until it reaches 100. The Node holds each closed batch until the next one closes, so that at the close of polls a short last batch joins the one before it; a Node that casts fewer than 100 ballots in the whole election anchors them as one batch after the polls close, the same crowd its precinct count already publishes. Each batch's sorted fingerprints are published with its root on the public verification site, so anyone can count a batch's ballots and rebuild its root. Token burns are anchored with their batch, never one by one, and batch times are published rounded to the hour. An anchor counts as final once it is more than 129,600 slots — Cardano's one-second units of time, so 36 hours — deep in the chain: the depth, $3k/f$ slots with $k = 2,160$ and $f = 0.05$ on mainnet, beyond which Cardano's documentation treats a transaction as immutable (Input Output Global, n.d.).

The sample batch on the cover is computed exactly this way. It holds eight ballots — far fewer than the 100 a real batch requires — so that every value fits on a page. Its inputs and every value are listed in the companion file *Figures: Worked Examples*, so any reader can check them.

FIGURE 11 · SECTION 8.1 · PART THREE

The reconciliation check, ballot 0515 0293

How the Automated Reconciliation Protocol compares the paper with the digital record

RECORDED BY THE VOTER TABLET

INDEX: 05150293
CONTEST 1: CHOICE A
CONTEST 2: CHOICE A
MEASURE 3: NO

R · 70 bytes of UTF-8 text, one line per field

THE PAPER · READ BY THE VOTER AND BY OCR

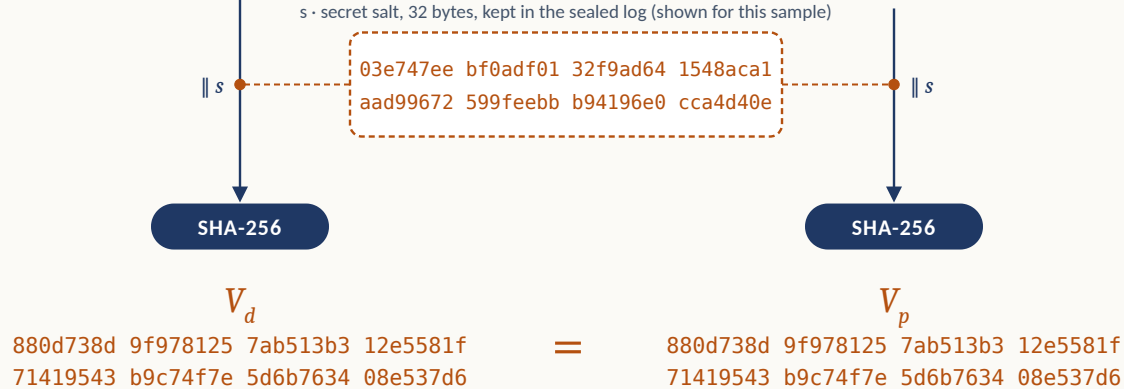
CONTEST 1
CHOICE A
CONTEST 2
CHOICE A
MEASURE 3
NO

0515
0293

INDEX: 05150293
CONTEST 1: CHOICE A
CONTEST 2: CHOICE A
MEASURE 3: NO

OCR

R rebuilt from the
printed characters



S = 1 · the paper and the digital record agree

WHAT IF THE TABLET HAD QUIETLY RECORDED SOMETHING ELSE

Recorded by the tablet
Printed, as the voter chose

CONTEST 2: CHOICE B
CONTEST 2: CHOICE A

$V'_d =$ 619d2d8c c9fa14b2 489dbe01 9a50f3e9 2b416d2c ba041ae3 69a17e44 2de1257d

$\neq V_p$, so $S = 0$. If caught before filing, the ballot is returned and the voter marks

a new one; if found after filing, the paper the voter read governs (Section 8.2, Figure 13).

$$V_d = V_p$$

$V_d = \text{SHA-256}(R_{\text{tablet}} \parallel s)$

$V_p = \text{SHA-256}(R_{\text{OCR}} \parallel s)$

$S = 1$ if they are equal, 0 if not

One ballot, read twice, fingerprinted twice.

If the fingerprints differ, the paper wins.

Both readings are fingerprinted with the same secret salt. Equal fingerprints mean the two agree character for character; change one character and the fingerprint no longer matches.

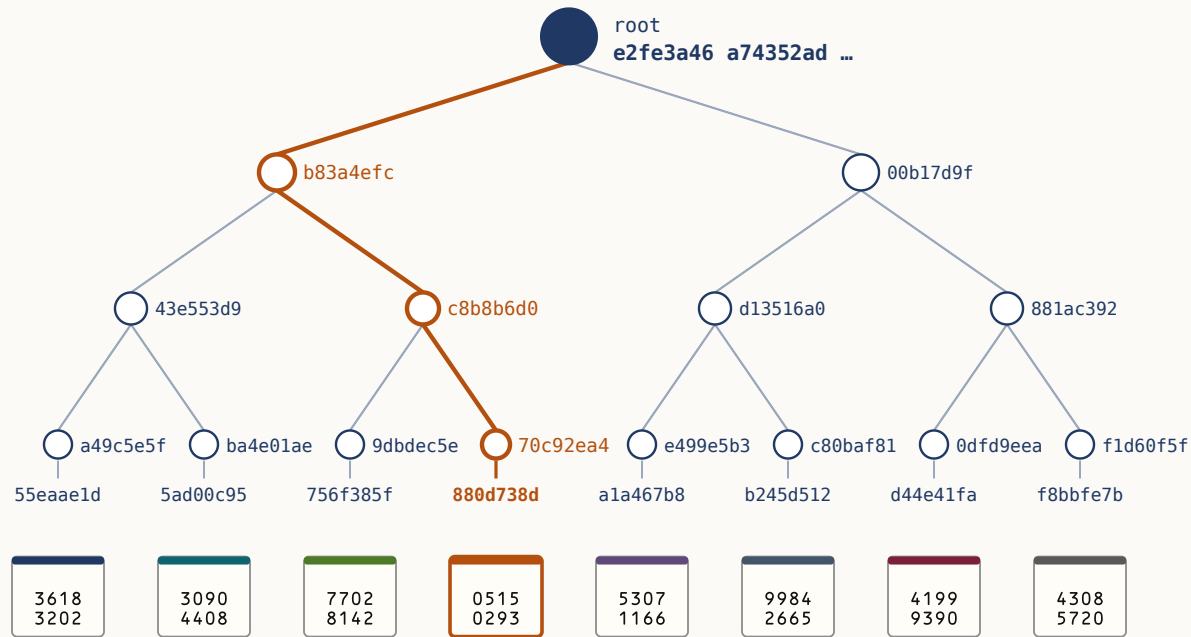
Check it: SHA-256 of the 70 bytes of R followed by the 32 bytes of s (102 bytes in all). Every byte is listed in *Figures: Worked Examples*.

FIGURE 12 · SECTION 8.1 · PART THREE

How a batch becomes one root

Batch 0003's eight fingerprints, sorted, then hashed pairwise per RFC 9162

Each circle shows the first 8 of its 64 hexadecimal digits.



Sorted in ascending order of fingerprint (55ea < 5ad0 < 756f < 880d < a1a4 < b245 < d44e < f8bb), so the tree says nothing about the order in which ballots were cast.

WHEN A BATCH CLOSES (SECTION 8.1)

60 min

a batch closes after an hour,

4,096

or at this many ballots,

≥ 100

but never with fewer;

held

until the next one closes

Each batch's sorted fingerprints are published with its root, so anyone can count its ballots and rebuild the root.

This sample batch holds 8 ballots so every value fits on the page; a real batch holds at least 100, except at the smallest Nodes.

$h = H(0x01 \parallel \text{left} \parallel \text{right})$

left, right the two hashes below a node; H SHA-256

leaf = SHA-256(0x00 $\parallel$ fingerprint)

8 leaves $\rightarrow 4 \rightarrow 2 \rightarrow 1$ root, 3 levels up

0x00 and 0x01 keep a node from passing as a ballot

***Eight fingerprints in,
one root out.***

The root is a fingerprint of the whole batch: change any ballot's record, or add or drop one, and the root changes. Only the root goes on Cardano.

Check it: hash each listed fingerprint to a leaf, then each pair upward; the last pair gives e2fe3a46...f8d72a75. 0293's path is orange: 70c92ea4 $\rightarrow$ c8b8b6d0 $\rightarrow$ b83a4efc $\rightarrow$ root.

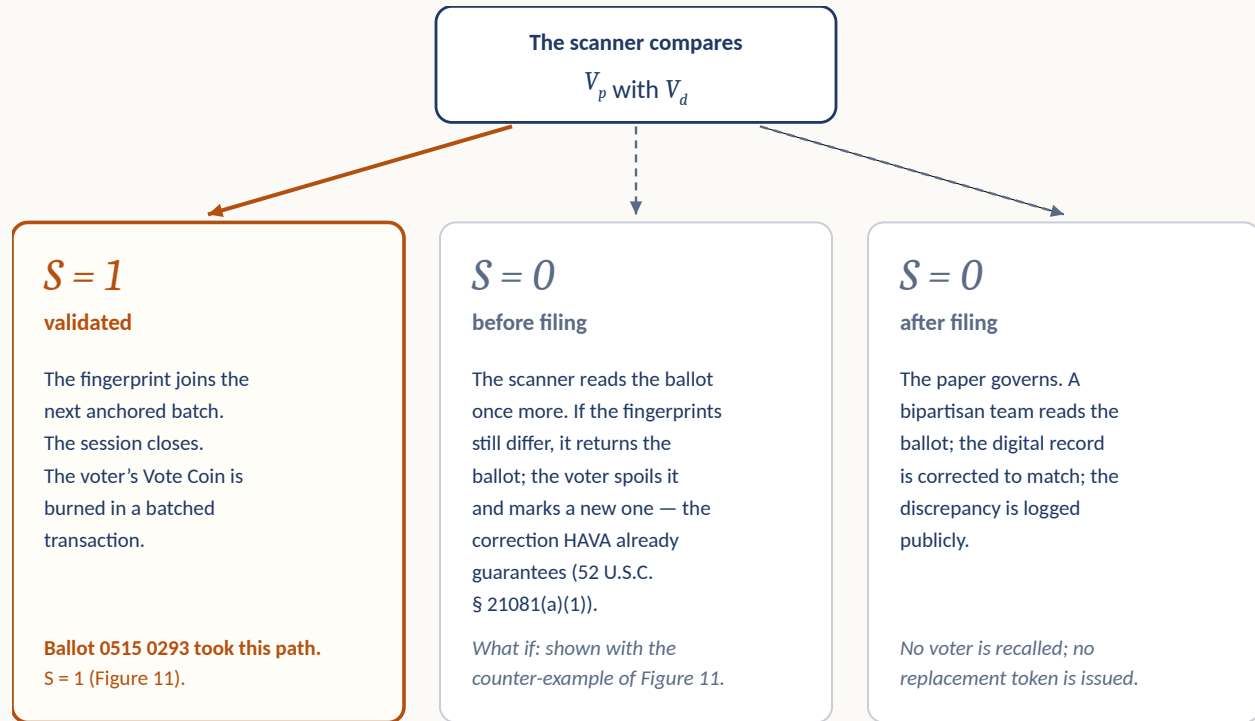
8.2 ARP Outcomes

- **S = 1 (validated).** The ballot's fingerprint joins the next anchored batch, the session closes, and the voter's Vote Coin is burned in a batched transaction (from Phase III, its credential tag is recorded in the batch instead — Section 12.1).
- **S = 0 before filing.** The scanner reads the ballot once more; if the fingerprints still differ, it returns the ballot, and the voter spoils it and marks a new one — the correction HAVA already guarantees (52 U.S.C. § 21081(a)(1)).
- **S = 0 after filing.** The paper governs. A bipartisan team reads the ballot, the digital record is corrected to match, and the discrepancy is logged publicly. No voter is recalled and no replacement token is issued.

FIGURE 13 · SECTION 8.2 · PART THREE

Three outcomes, one rule

What happens after the reconciliation check, for a match and for each kind of mismatch



THE ONE RULE UNDER ALL THREE

The paper the voter read is the ballot of record.

A discrepancy never changes a vote on the voter's behalf: before filing, the voter decides; after filing, the paper decides. Discrepancy rates are published on the public dashboard within 72 hours of poll close.

Phase I passes only with zero unexplained discrepancies in at least 300,000 reconciled ballots (Figure 23).

100% *checked*

every tablet-printed ballot, as it is deposited — not a sample, weeks later

$S = 1$ match: the ballot goes on

$S = 0$ mismatch: the paper governs

Before filing, the voter corrects.

After filing, the paper decides.

The three outcomes of Section 8.2. Every one ends with the paper ballot intact in the archive, and every discrepancy is logged where the public can count it.

Check it: for 0293, $V_d = V_p = 880d738d...08e537d6$, so $S = 1$; the counter-example's $619d2d8c...2de1257d$ differs, so $S = 0$.

PART FOUR

Privacy & Law

Sections IX – X · Privacy Impact Assessment · Legal, Regulatory & Compliance Framework

WHO THIS PART IS FOR

Attorneys, privacy officers, compliance staff, and civil-liberties reviewers — especially anyone scrutinizing the biometric and identity provisions.

IN THIS PART

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IX. Privacy Impact Assessment (PIA)

A Privacy Impact Assessment is the structured analysis federal law (E-Government Act of 2002, Section 208) requires for systems that handle personal information. The inventory below covers every data element the protocol touches.

9.1 Data Inventory

Data Element	Classification	Storage Location	Retention & Disposal
DID / voter identifier	PII (personally identifiable information) — sensitive	Voter's credential card; issuance record held by the jurisdiction	Card persists until revoked; issuance records kept per state records law
Biometric template	PII — biometric	Voter's credential card (opt-in only); read into the Secure Node for the matching session	On the card until the voter revokes it; destroyed in the Node at session end, never written to disk or transmitted
Vote Coin assignment	Pseudonymous personal data (linkable only by the issuing jurisdiction)	Cardano ledger (public, pseudonymous)	Permanent on-chain record, pseudonymous only
Ballot preference (candidate choice)	Sensitive — secret	Physical ballot (anonymized) and the Node's encrypted vote package; never published, never on-chain	Retained per ballot-retention law (typically 22 months federal)
Ballot fingerprint and salt (ARP record)	Non-PII (cryptographic)	Secure Node sealed audit log, encrypted so that decrypting it needs key shares held by all three Triad institutions; each batch's sorted fingerprints, never the salts, are published on the public verification site, and only batch roots are anchored on Cardano	Sealed log kept with the paper ballots for the retention period; published fingerprints and anchored roots permanent

 **What never touches the blockchain:** No ballot choice, no name, no biometric, and no personally identifying information is ever written to the chain — only Merkle roots of salted fingerprints, and pseudonymous token events. Fingerprints are salted with secret random values, sorted before each batch is built, and anchored with coarsened timestamps, so an observer — including one with a future quantum computer — learns that ballots were cast and reconciled, not how anyone voted.

FIGURE 14 · SECTION IX · PART FOUR

Public, sealed, never recorded

The Section 9.1 data inventory, and who can ever see each item

Public	Sealed	Never recorded
<i>anyone can see and check it</i>	<i>kept, never published</i>	<i>it exists nowhere</i>
Batch roots, on Cardano e2fe3a46...f8d72a75 Each batch's sorted fingerprints, on the verification site 55eaae1d 5ad00c95 756f385f 880d738d ← 0293 a1a467b8 ... 8 in all	Salts, in the sealed audit log 03e747ee... ← 0293 <i>shown only because this is a sample</i> The paper ballots, anonymized, in batch boxes  kept for the legal retention period (22 months, federal)	Any link from a voter to a ballot or its index  no log, time or sequence The check-in time of any ballot Casting order within a batch the box is by index, the tree by fingerprint: neither is time The face data, after the session destroyed in the Node at session end; never on disk
Token events, pseudonymous issue and burn of each Vote Coin; only the jurisdiction can link them Batch times, to the hour batch 0003: 09:00–10:00 Counts and rates by precinct, on the dashboard within 72 hours of poll close	Card issuance records held by the jurisdiction A face template, if chosen on the voter's own card only; no central face database Opening the archive or the log needs all three keys Vote packages, encrypted, AES-256, inside the Node only	

The chain holds only roots and pseudonymous token events. A future computer that reads all of it learns that ballots were cast and reconciled — not how anyone voted: no ballot content is ever written to the chain.

Unlinkability is built in: a link the Node never records cannot later be leaked or demanded, even by the three Triad institutions acting together.

2^{256} possible salts

a fingerprint alone reveals nothing: to test a guessed ballot, one would also need its 32-byte salt — one of about 1.16×10^{77} values, fresh for every ballot

*Public enough to check.
Sealed enough to keep the secret.*

The data inventory of Section 9.1 in three columns. Publishing each batch's sorted fingerprints lets anyone count its ballots and rebuild its root; the salts that would give them meaning are never published.

Check it: the public list for batch 0003 has 8 fingerprints and the box holds 8 ballots; the root rebuilt from the list alone is e2fe3a46...f8d72a75 — no salt needed.

9.2 Privacy Principles

- **Data minimization.** The system collects only what identity verification and vote recording strictly require. No location history, browsing behavior, or metadata beyond a coarsened batch timestamp is retained.
- **Purpose limitation.** Biometric data is used solely for authentication at the moment of ballot issuance. It is never shared with any third party, law-enforcement agency, or federal database.
- **Unlinkability.** The zero-knowledge architecture ensures that even the three Triad institutions acting together cannot connect a voter's identity to their ballot choice after the session closes.
- **Nothing to hand over.** The Node keeps no record linking a voter, session, or check-in time to a ballot or its validation index. It retains exactly what federal and state law require, and none of it can connect a voter to a vote.
- **Informed consent.** Biometric capture is optional. A voter who chooses it consents explicitly at enrollment, after a plain-language explanation of use, retention, and destruction; every other voter uses the card and PIN, and in-person verification by election officials is always available.
- **Access and correction.** Voters may review their DID record and transaction status at any Secure Node during the post-election audit window, per the Voter Bill of Rights (Section XVII).

X. Legal, Regulatory & Compliance Framework

10.1 Federal Statutory Compliance

Statute / Standard	How the Protocol Complies
Help America Vote Act (HAVA) 2002, §301	Provides a voter-verifiable paper ballot; supports manual audit and recount; provides language access and error notification; preserves the provisional-ballot safety net of §302
VVSG 2.0 (EAC, 2021)	Software-independent by design — the paper record, not software, determines the outcome; designed against all fifteen VVSG 2.0 principles and tracked against the draft VVSG 2.1
Executive Order 14248 §4(b); draft VVSG 2.1 (September 2025 revision) Req. 3.3-C and 3.3-D	No voter selection is encoded in any barcode or QR code. Selections are printed as human-readable text and read by OCR; the only other machine-read element is a human-readable validation index, a non-selection identifier the draft expressly permits
Voter eligibility (state registration law; Executive Order 14399 and the SAVE America Act, as of September 2026)	Eligibility comes from the state's voter registration database, however the state maintains it — including any citizenship verification federal law or policy requires (as of September 2026, Executive Order 14399, whose citizenship-list provisions the Supreme Court let take effect pending appeal, and the SAVE America Act, passed by the House and pending in the Senate). The protocol takes no position on either. Mail ballots stay outside the protocol, under state law.
Americans with Disabilities Act (ADA)	WCAG 2.1 AA interfaces (the ADA Title II standard), targeting WCAG 2.2 AA; haptic and audio guidance; accessible kiosk height and reach ranges

Statute / Standard	How the Protocol Complies
Voting Rights Act, §203	Multi-language ballots for covered language-minority groups
18 U.S.C. §§ 592–593	No troops, armed personnel, or military interference at any polling place: every technical certification is completed before voting opens and away from the polling place, and the reference model's key-holders are civilian
E-Government Act of 2002, §208	Privacy Impact Assessment conducted and included (Section IX); PII handling fully documented
FIPS 140-3	Cryptographic modules are FIPS 140-3 validated. VVSG 2.0 still names FIPS 140-2, but NIST's transition schedule places every 140-2 certificate on its Historical List as of September 22, 2026 (NIST, 2026), and the EAC's Decision on Request for Interpretation 2026-01 (June 23, 2026) reads the "current FIPS 140 validation" that Requirement 13.3-A demands as an Active module only: Historical 140-2 modules are accepted case by case for systems submitted before January 1, 2027, and not at all after (EAC, 2026d). A FIPS 140-3 module meets the rule in every period. Only end-to-end verification functions use the exception VVSG 2.0 provides for them.
NIST SP 800-53 Rev. 5	Controls mapped across Access Control (AC), Audit (AU), Identification & Authentication (IA), System Integrity (SI), and Physical & Environmental (PE) families
Environmental standards	Cardano's Proof-of-Stake consensus consumes over 99% less energy than Proof-of-Work chains (Platt et al., 2021); solar arrays reduce grid dependency at each Node

10.2 The Certification Pathway

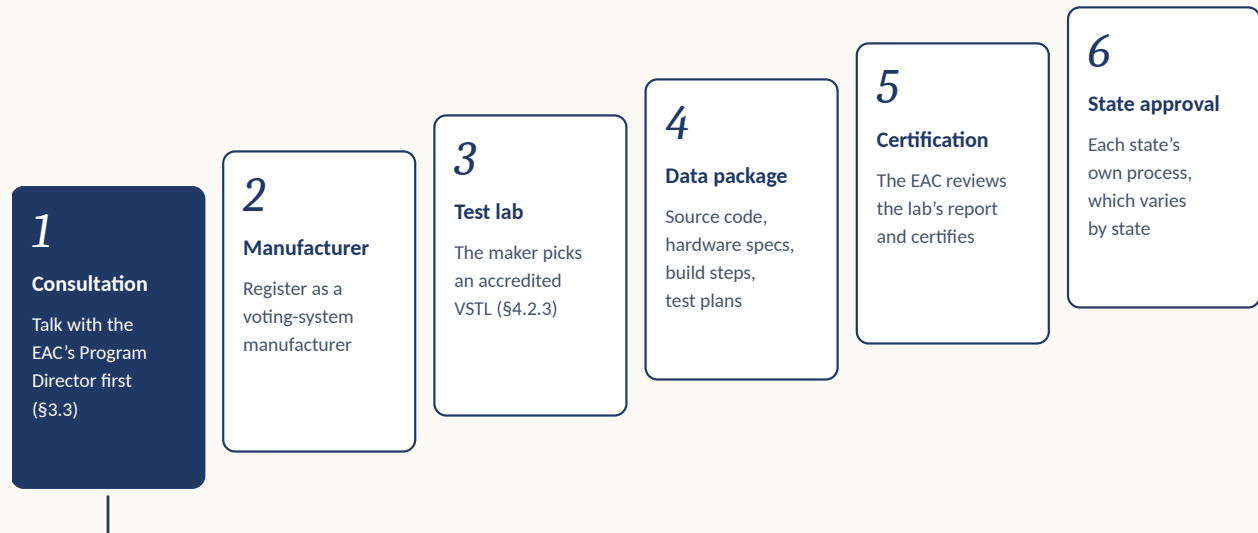
Federal EAC certification under VVSG 2.0 is the gateway to state-level procurement. The protocol follows the standard EAC Testing and Certification pathway:

1. **Early consultation.** Discuss the new technology with the EAC's Testing and Certification Program Director as early as possible (Program Manual §3.3).
2. **Manufacturer registration.** Register with the EAC as a voting-system manufacturer.
3. **VSTL engagement.** Select and contract an EAC-accredited Voting System Test Laboratory (VSTL) — an independent lab authorized to test voting equipment against federal standards — for conformance testing; the manufacturer, not the EAC, chooses the lab (Program Manual §4.2.3).
4. **Technical Data Package (TDP).** Deliver complete source code, hardware specifications, build instructions, and test plans.
5. **EAC review and certification.** On successful VSTL testing, the EAC reviews the report and issues federal certification.
6. **State submission.** Use federal certification as the foundation for individual state approval processes, which vary by jurisdiction.

FIGURE 15 · SECTION 10.2 · PART FOUR

From consultation to certification

The six steps of the EAC pathway, and where this protocol stands on them



This paper asks for step 1 (Section 18.2).

WHERE THE PILOTS FIT, AND THE LATER GATES (SECTION XVI)

Phases I–II	supervised pilots under the host state's authority for experimental equipment (e.g., Tenn. Code Ann. § 2-9-110)
Phase III gate	EAC certification, or the state's equivalent, and state election authority approval
Phase IV gate	EAC federal certification, including the penetration testing federal law now requires (52 U.S.C. § 20971(e))

The EAC certifies systems, not papers: a registered manufacturer, an accredited lab and a full data package come first.

6 steps on the pathway

the standard EAC Testing and Certification pathway, as Section 10.2 lists it
step 1 is a conversation, not an application

The first step is a conversation, not an application.

The EAC's Program Manual has a process for new technology: an early consultation before any application. Pilots under state authority test the design; their results go to the EAC before any application.

Check it: Program Manual v3.0, §3.3 (early consultation) and §4.2.3 (the manufacturer, not the EAC, chooses the lab) — both cited in Section 10.2.

PART FIVE

Operations & Scale

Sections XI – XIV · Comparative Analysis · Scalability · Onboarding · Implementation & Token Lifecycle

WHO THIS PART IS FOR

Election administrators, county clerks, budget and procurement officers, and logistics planners deciding whether this can actually be run on a Tuesday in November.

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XI. Comparative Analysis

The table below compares the protocol against the system families in current or proposed use: DRE machines (Direct-Recording Electronic — touchscreen machines that store votes only in memory), optical-scan paper systems, ballot-marking devices (BMDs — touchscreens that print a paper ballot for the voter to check, as this protocol's tablets do), end-to-end-verifiable (E2E-V) systems (Benaloh et al., 2015) such as ElectionGuard, which gave voters privacy-preserving tracking codes in real U.S. elections (Microsoft Democracy Forward et al., 2023), and internet voting.

Feature	Vote Coin Protocol	Traditional DRE	Optical Scan	Ballot-Marking Device	E2E-Verifiable (ElectionGuard)	Internet Voting
Voter-verifiable paper record	Yes — mandatory	Varies	Yes	Yes (printed ballot)	Yes, as deployed with paper	No
Air-gapped operation	Yes	Partial	Yes	Typically	Depends on host system	No
Cryptographic public audit trail	Yes (Cardano)	No	No	No	Yes (published encrypted records)	Varies
Ballot-secrecy-preserving receipts	Yes (tracking code + inclusion proof)	No	No	No	Yes (tracking codes)	Rare
Multi-party governance (three keys)	Yes	No	No	No	Partial (key shared among guardians)	No
Per-ballot physical-digital reconciliation	Every ballot, continuously	No	Sampled (ballot-level comparison audits)	Sampled	Encrypted-record checks by voters and observers	No
VVSG 2.0 software independence	Yes (paper is the record; see Objection 7)	Varies	Yes	Only if voters check the paper	Yes (end-to-end verification)	No
Remote attack surface	None (air-gapped)	Limited	Minimal	Minimal	Depends on host system	Very high

The nearest existing cousin is optical scan with risk-limiting audits — a genuinely good system. The protocol's claim is not that optical scan is broken, but that periodic, sampled auditing can be upgraded to *continuous, per-ballot, publicly verifiable* auditing without sacrificing anything optical scan does well. Section XIX, Objection 50, addresses this comparison head-on.

XII. Scalability & Performance

12.1 The Design Principle: Local-First, Chain-Anchored

A common and fair criticism of blockchain voting concepts is throughput. Cardano's main chain carries roughly 4.5 kilobytes of transactions per second — about 3.5 transactions per second observed on mainnet (Input Output Global, 2026) — while the 2024 general election counted more than 158 million ballots, cast over weeks: over 72% in person, on or before Election Day, and about 30% by mail (EAC, 2025a). The protocol resolves this not by hoping the chain gets faster, but by refusing to put the chain in the critical path of casting a vote.

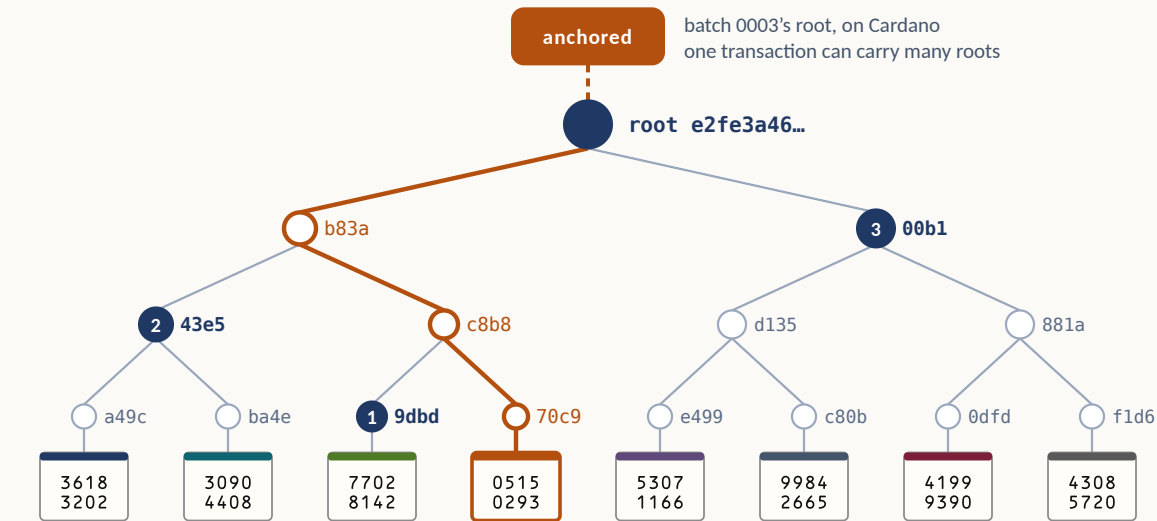
- **Casting is local and instant.** A voter's ballot is complete the moment the paper is verified and deposited. The Secure Node records the vote package locally on encrypted, FIPS-validated storage. No voter ever waits on a network.
- **Anchoring is asynchronous and batched.** Ballot fingerprints are assembled into a **Merkle tree** — a cryptographic structure that condenses thousands of records into a single fingerprint, while still letting any individual record prove its membership — and the tree's root (built as specified in Section 8.1) is committed to Cardano at regular intervals. One on-chain transaction can anchor an entire precinct's activity, and each voter's receipt carries a tracking code whose **inclusion proof**, published once the batch is anchored, ties the ballot's fingerprint to that public anchor.
- **Chain limits, as of September 2026.** Cardano's protocol parameters allow one block of up to 90,112 bytes about every 20 seconds on average — about 4.5 kilobytes a second — and a single transaction of up to 16,384 bytes, while each batch root is 32 bytes. Cardano's on-chain governance can change these limits, so check the current values when applying this design (Cardano mainnet protocol parameters, 2026; Input Output Global, n.d.).
- **Burns scale with voters, not Nodes.** Roots are one 32-byte value per Node per hour. Each card voter's Vote Coin, by contrast, is its own on-chain output, burned by spending it, so the burn load grows with voters. At the 2024 general election's in-person volume — over 72% of more than 158 million ballots, about 114 million — one burn per card voter at a generous 100 bytes of transaction data each (a batched burn adds about 57 bytes) comes to about 11.4 gigabytes: about 29 days of the entire chain's September 2026 capacity, before anyone else's traffic (the arithmetic is in the reader kit published with this paper). That is the number a reader of Park, Specter, Narula and Rivest (2021) will look for, so it is stated here rather than hidden. The design's answer: per-voter Vote Coins on the main chain serve Phases I–II, where the table below shows the load is small; from Phase III, each batch record instead lists the **credential tags** — the pseudonymous identifier each voter's right carries (Section 14.1) — spent in that batch, hashed into its root, and a tag may appear in only one anchored batch, which anyone can check across the published batch lists. That is the same one-right-one-use check, at one transaction per batch, and it is what keeps the design inside the chain's budget at state and national scale. This paper describes the token as it runs in Phases I and II; everything said of a burn holds for a spent tag from Phase III.
- **The canvass never waits on the chain.** Unofficial precinct totals are read from the paper on election night, as they are today, and the official canvass follows the host state's statute and calendar. Chain anchors are an audit index: an anchor that arrives late, or a batch whose anchor is delayed by a network outage, is a logged exception, not a reason to hold a count.

- **Layer-2 headroom.** If more headroom is ever wanted, Hydra — Cardano’s Layer-2 scaling framework, which processes transactions in fast local channels and settles their final state to the main chain — can provide per-Node-cluster channels with sub-second finality (average confirmation times of about 5 milliseconds for one node and 28 for three in the project’s published end-to-end benchmarks, run on cloud machines; Input Output Global, 2025). Hydra reached its production-ready 1.0 release in October 2025 (Input Output Global, 2025); Ouroboros Leios, which targets far higher main-chain throughput, is targeted for testnet in 2026 (Input Output Global, 2026). The design assumes neither: one anchoring transaction can carry the signed roots of many Nodes.

FIGURE 16 · SECTION 12.1 · PART FIVE

Proving ballot 0515 0293 was anchored

How the receipt's tracking code and a Merkle inclusion proof reach the public anchor



Receipt • tracking code

880d738d...
...08e537d6

The code is the ballot's own fingerprint — never its index. Once batch 0003 is anchored, the voter or anyone fetches the proof for that code from the public verification site and checks it — three hashes, three steps:

Start SHA-256(0x00 || 880d738d...08e537d6)
 = 70c92ea4...ce8bd14c

1 sibling on the left SHA-256(0x01 || 9dbdec5e...6e14c705 || 70c92ea4...ce8bd14c)
 = c8b8b6d0...e03f1d54

2 sibling on the left SHA-256(0x01 || 43e553d9...86826a9f || c8b8b6d0...e03f1d54)
 = b83a4efc...bc639c22

3 sibling on the right SHA-256(0x01 || b83a4efc...bc639c22 || 00b17d9f...b24667b8)
 = e2fe3a46...f8d72a75
 = the root anchored on Cardano, e2fe3a46...f8d72a75 ✓

$$k = \lceil \log_2 n \rceil$$

at most k hashes prove one of n ballots; 32 bytes each

$n = 8$ (this sample) $\rightarrow$ 3 hashes, 96 bytes

$n = 500$ (a Phase I peak hour) $\rightarrow$ 9 hashes, 288 bytes

$n = 4,096$ (the batch cap) $\rightarrow$ 12 hashes, 384 bytes

*Three hashes prove this ballot is in the batch.
At most nine would, in a Phase I peak hour.*

The proof shows membership only: nothing the published list does not already show, and — because the fingerprint is salted and the salt stays sealed — nothing about what 0293 says. At most $\lceil \log_2 n \rceil$ hashes are needed, whatever the batch.

Check it: leaf = SHA-256(0x00 || fingerprint), node = SHA-256(0x01 || left || right) — IETF RFC 9162, §2.1.1. The sides (left, left, right) follow from sorting the fingerprints in ascending order.

12.2 Phase-by-Phase Digital Throughput

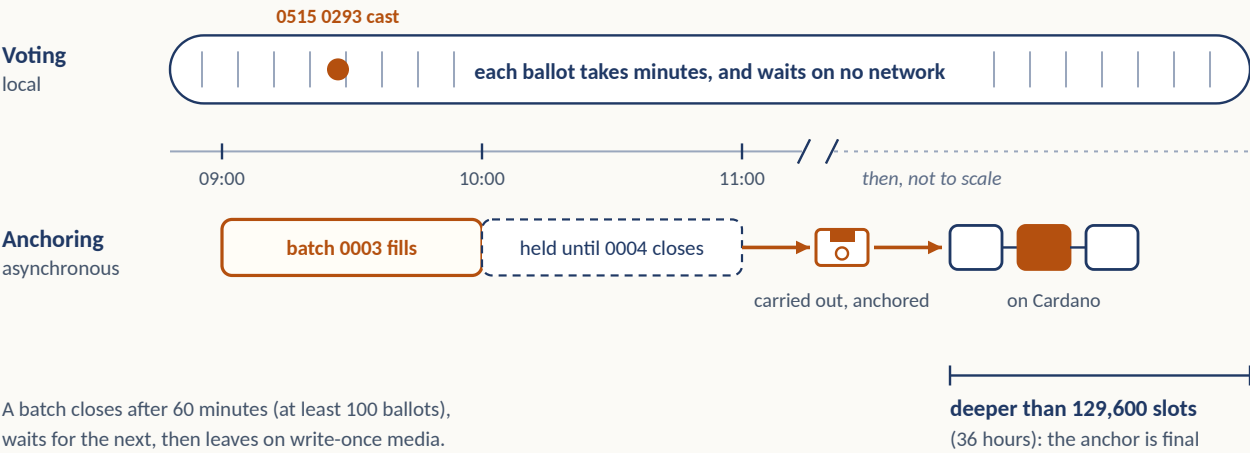
- **Municipal (Phase I).** Peak $\approx$ 200–500 votes/hour per Node. Direct mainnet anchoring; no Layer-2 required.
- **County (Phase II).** Peak $\approx$ 2,000–5,000 votes/hour county-wide. Batched Merkle anchoring on mainnet; Hydra optional.
- **State (Phase III).** Peak up to 50,000 votes/hour statewide. Batched anchoring on mainnet, many Nodes' signed roots per transaction; from this phase, spent credential tags travel in the batch record instead of per-voter burns (Section 12.1); Hydra optional.
- **National (Phase IV).** Full Election Day load. The same batched anchoring and per-batch tag lists, with automatic fallback to local encrypted storage during any congestion event — voting never pauses for the network. Hydra remains optional headroom.

FIGURE 17 · SECTIONS 12.1–12.2 · PART FIVE

Local first, anchored later

Why the chain’s speed never slows a voter, and when an anchor becomes final

TWO CLOCKS, ONE NODE (SAMPLE TIMES)



A batch closes after 60 minutes (at least 100 ballots), waits for the next, then leaves on write-once media.

If Cardano is congested or down, only the anchor waits — and the Node keeps every signed batch, so nothing is lost.

THE LOAD THE CHAIN ACTUALLY SEES

Phase I • a municipal Node	200–500 an hour at peak	A Node closes at most one batch an hour (it prints at most 1,500 an hour, under the 4,096 cap), so the chain sees one root per Node per hour — plus, through Phase II, one Vote Coin burn per card voter, sent with the batch; one transaction carries many roots.
Phase II • a county	2,000–5,000 an hour at peak	
Phase III • a state	up to 50,000 an hour at peak	

About 3.5 transactions a second are observed on Cardano’s main chain — 12,600 an hour (Input Output Global, 2026).

The chain’s speed never limits voting, because the protocol never puts the chain in the voter’s path.

$$3k / f = 129,600$$

$k = 2,160$, Cardano’s security parameter
 $f = 0.05$, the active-slot coefficient
slots of 1 second: 129,600 $s = 36$ hours
mainnet genesis values; the depth Cardano’s documentation treats as immutable

*Voting happens now.
The public anchor follows.*

Section 8.1’s batch rules on one timeline. The last batch closes with the polls; its anchor is final once it is more than 129,600 slots (36 hours) deep. From Phase III, spent tags ride in the batch record, not per-voter burns (Section 12.1).

Check it: $3 \times 2,160 \div 0.05 = 129,600$ slots; $\div 3,600 = 36$ hours; and $3.5 \times 3,600 = 12,600$ transactions an hour.

The table puts the two loads side by side at the September 2026 limits (Section 12.1), for the peak hour of each phase. The batch roots never matter; the per-voter burns are what force the Phase III path.

Phase	Peak load	Batch roots on the chain	Per-voter burns at 100 bytes each (chain carries 16.2 MB an hour)	Path
Municipal (Phase I)	500 ballots an hour	32 bytes per Node per hour; 1,000 Nodes are 32 kB, 0.2% of the chain	50 kB an hour, 0.3% of the chain	Per-voter Vote Coins on the main chain
County (Phase II)	5,000 ballots an hour	32 bytes per Node per hour; 1,000 Nodes are 32 kB, 0.2% of the chain	500 kB an hour, 3.1% of the chain	Per-voter Vote Coins on the main chain
State (Phase III)	50,000 ballots an hour	32 bytes per Node per hour; 1,000 Nodes are 32 kB, 0.2% of the chain	5,000 kB an hour, 30.8% of the chain	Spent credential tags in each batch record; roots only on the chain
National (Phase IV)	2024: about 114 million in-person ballots over the election period	As above: one root per Node per hour	11.4 GB in all — about 29 days of the whole chain	Spent credential tags in each batch record; roots only on the chain

12.3 Physical Throughput

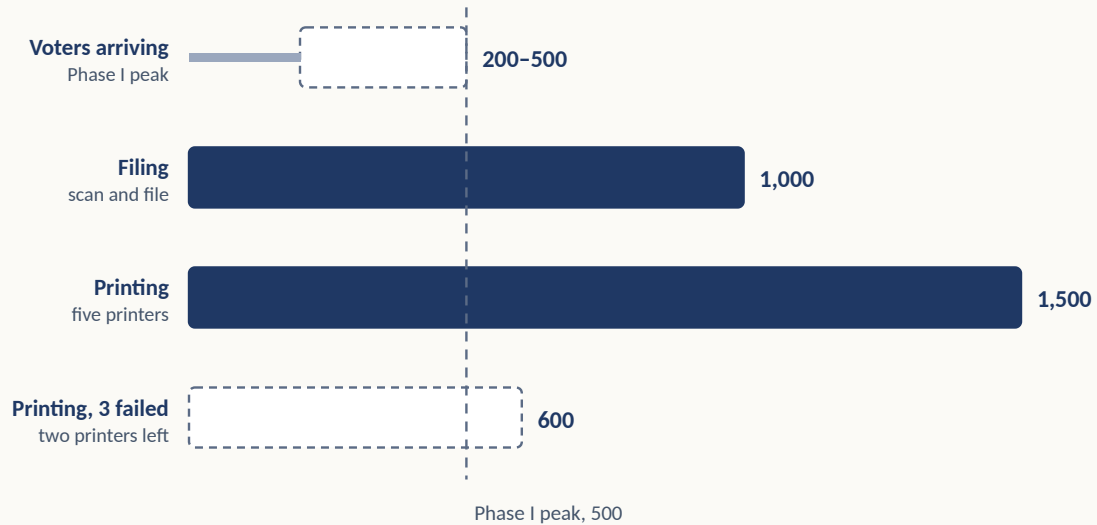
- **Printer array capacity.** Each archival-grade printer produces roughly 300 ballots/hour; five in active-active configuration yield up to 1,500/hour per Node, with any single failure costing 20% capacity rather than any downtime.
- **Robotic filing speed.** Filing systems are specified at 1,000 scans-and-files per hour in Phase I stress testing — twice the Phase I peak of 500 ballots per hour per Node, so no physical backlog can accumulate.
- **Queue management.** Real-time queue telemetry displays wait estimates and can trigger deployment of additional Secure Nodes from a mobile reserve fleet when throughput falls below target.

FIGURE 18 · SECTION 12.3 · PART FIVE

Capacity per Node

What one Secure Node can print, check and file in an hour, against the Phase I peak

BALLOTS PER HOUR, PER NODE



Of the machine steps, the slowest sets the pace:

filing, at 1,000 an hour — twice the Phase I peak. Printing runs at three times it.

The Phase I stress test

1,000 prints, scans and files an hour, sustained for 8 hours: 8,000 ballots through one Node (Section XVI).

When lines grow anyway

Queue telemetry shows wait estimates in real time, and a mobile reserve fleet of Secure Nodes deploys where throughput falls below target. Nodes are sized to precinct population, like any polling place.

Nothing here waits on a network: these are the machine limits, and each is set above the peak.

$$T = \min(P, F) = 1,000$$

P printing: $5 \times 300 = 1,500$ an hour

F filing: 1,000 an hour (Section 12.3)

T what one Node's printers and filing can finish, per hour

$T \div 500 = 2\times$ the Phase I peak

Filing is the slowest machine step, and it still runs at twice the peak.

Rates from Sections 12.2–12.3. With three printers down, printing (600 an hour) becomes the slowest step, and it still exceeds the Phase I peak of 500.

Check it: $\min(1,500, 1,000) = 1,000$; $1,000 \div 500 = 2$; $1,500 \div 500 = 3$; $1,000 \times 8 = 8,000$.

XIII. Onboarding, Accessibility & Voter Education

13.1 Digital Identity Enrollment

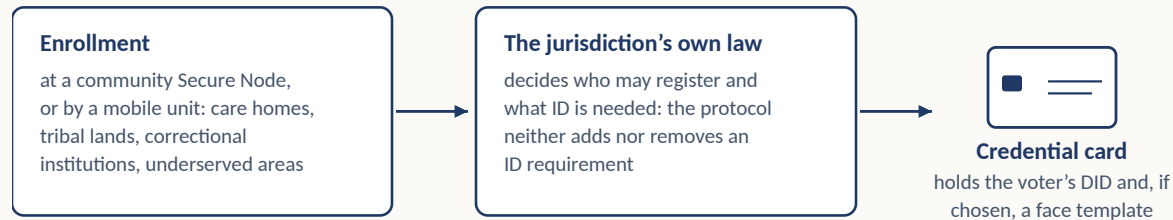
- **Identus DID setup.** Citizens enroll at community Secure Nodes during a pre-election window. Enrollment applies the jurisdiction's existing registration and identification law — the protocol neither adds nor removes an ID requirement. Biometric capture is voluntary, and a non-biometric path is always available. Eligibility comes from the state's voter registration database, not from the credential: a voter whose card fails on Election Day uses the ordinary poll-book check-in (Pillar 1; Objection 35), so the Identus layer is never a condition of the right to a paper ballot.
- **Assisted enrollment.** Trained election officials assist in person, and a mobile enrollment program travels to underserved communities, care homes, and correctional facilities to guarantee equitable access.
- **Portability.** The voter carries a credential card — a smart card holding the voter's decentralized identifier and, if the voter chose, a face template. The card persists across elections; the jurisdiction's issuance records follow state records law. Re-enrollment is never required — only the card and PIN at the moment of voting.

FIGURE 19 · SECTION 13.1 · PART FIVE

Enroll once, vote with a card

How a voter gets a credential card, and the three ways in at the polls

ENROLL ONCE

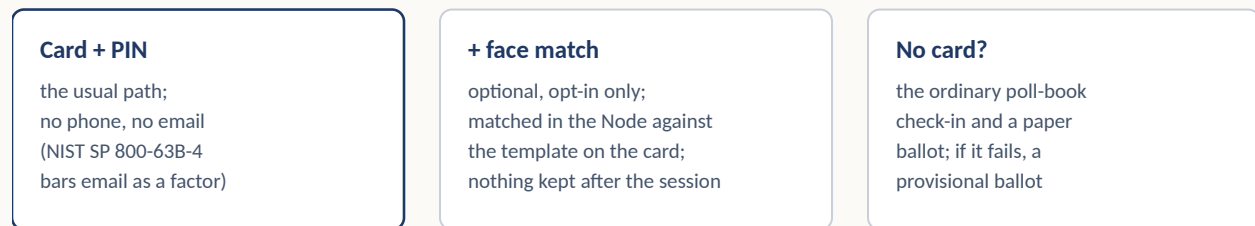


THEN, AT EVERY ELECTION AFTER



The card persists; re-enrollment is never required. Only the card and a PIN are needed at the moment of voting.

AT THE POLLS, THREE WAYS IN — NONE TURNS A VOTER AWAY



Provisional ballots follow HAVA §302 and are adjudicated under existing state procedures: technology failure can never subtract a voter.

Face matching, where chosen, uses only algorithms benchmarked in NIST's FRTE with a false-match rate of 1 in 10,000 or better for every demographic group, and presentation-attack detection (Objection 22).

1 enrollment

for **every election** after; **free**, and it travels **card + PIN** at the polls; face match optional **no card** never means no vote

Enroll once. Vote with a card and a PIN.

Section 13.1. The card holds no ballot information and never connects to one: identity decides who may vote, and stops there (Figure 2).

Check it: against the text — Pillar 1 (card, PIN, optional face match, poll-book fallback), Section 13.1 (enrollment, portability) and Objection 35 (HAVA §302 provisional ballots).

13.2 Post-Election Forensics & Spot-Check Audits

- **Random batch audit.** Any accredited auditor may pick any anchored batch from the election period; the robotic archive retrieves the corresponding physical ballots within seconds for spot verification.
- **Risk-limiting audits.** The physical archive is structured to support RLAs, which VVSG 2.0 Principle 9 (Auditable) requires systems to support, with sampling methods from the statement on risk-limiting audits by the American Statistical Association (2010) and the published literature (Lindeman & Stark, 2012).
- **Public transparency dashboard.** Aggregated, anonymized statistics — counts by precinct, ARP discrepancy rates, audit completion — are published within 72 hours of poll close.

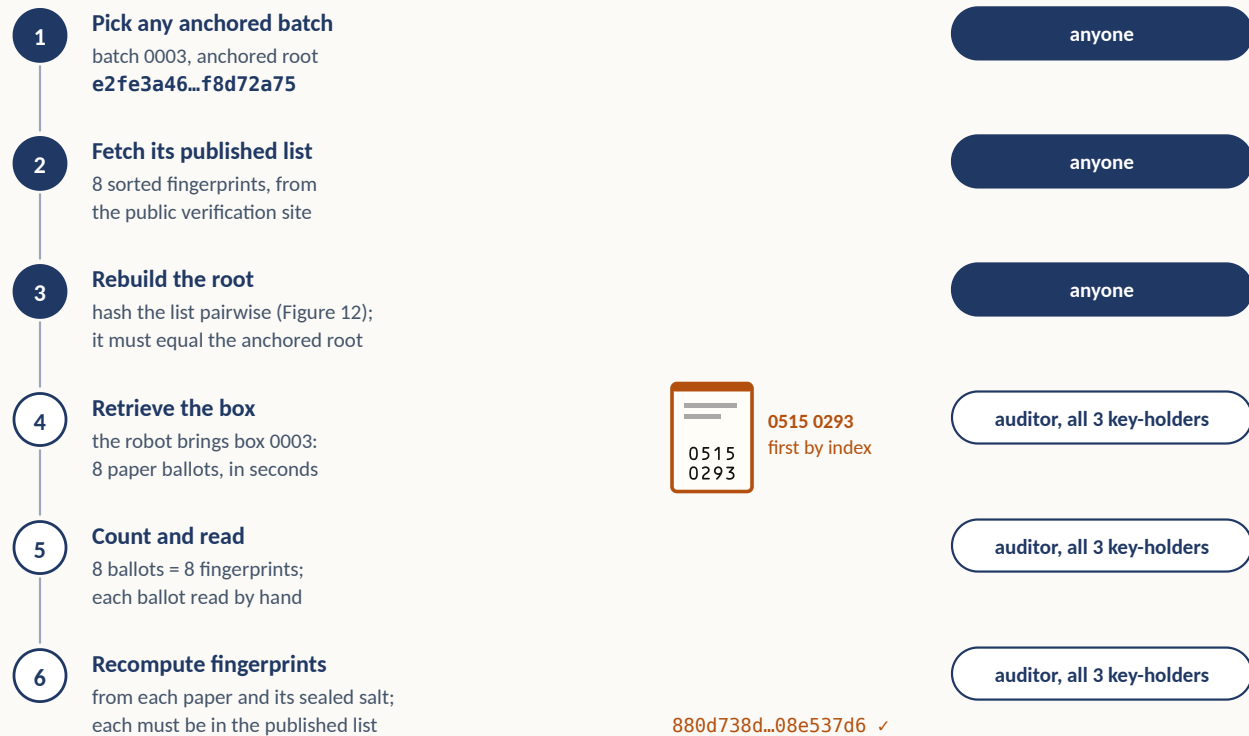
FIGURE 20 · SECTION 13.2 · PART FIVE

Audit any batch

What anyone can check from public data, and what an auditor checks against the paper

AUDITING ONE BATCH

WHO CAN DO IT



Risk-limiting audits use the same archive: a statistical sample of paper ballots large enough to confirm, to a chosen confidence, that the reported winner won (Lindeman & Stark, 2012); VVSG 2.0 Principle 9 requires support for them.

Steps 1–3 need no permission at all; steps 4–6 open the archive, so they follow the Three-Key Rule.

$$|\text{box}| = |\text{list}| = 8$$

|box| paper ballots in the batch's box
 |list| fingerprints published for the batch
 and the list, hashed pairwise, must give
 the anchored root

*Any anchored batch:
 count it, hash it, read it.*

Section 13.2's random batch audit. Each check can fail on its own: a missing ballot, an extra fingerprint, or a changed record each breaks one of the equalities.

Check it: batch 0003's list has 8 entries; hashed pairwise it gives e2fe3a46...f8d72a75, the anchored root; 0293's entry is 880d738d...08e537d6.

XIV. Implementation & Token Lifecycle

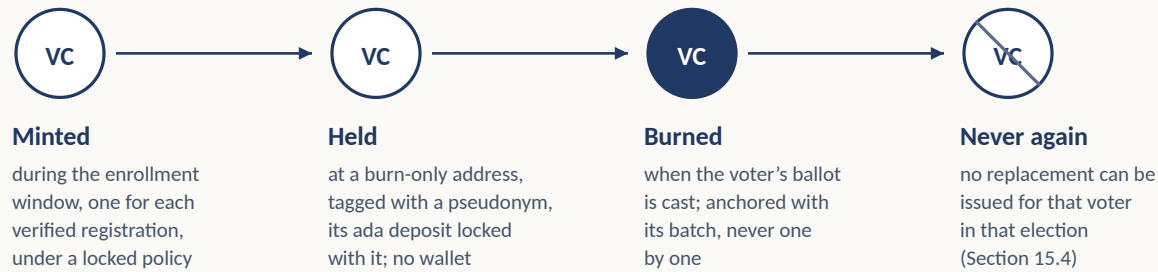
14.1 Vote Coin Issuance

- **Minting.** Vote Coin is minted as a Cardano **native asset** — a token type the ledger itself tracks and enforces (Input Output Global, n.d.) — under a locked minting policy, in batched transactions during the enrollment window.
- **Distribution logic.** Issuance is strictly 1:1 against the Voter Registration Database, during the pre-election enrollment window, gated on successful Identus DID verification.
- **Non-transferability.** Each Vote Coin is held at a burn-only validator address, tagged with a pseudonymous identifier for its voter — its credential tag, the value each batch record lists from Phase III (Section 12.1); it never enters any wallet, so there is nothing to transfer (Section 4.2).
- **Non-reusability.** Once spent, no replacement can be issued for the same cycle (Section 15.4).

FIGURE 21 · SECTION 14.1 · PART FIVE

A Vote Coin's life

From minting to a single burn, and the count that must balance at every Node



The Vote Coin belongs to the identity side: navy, pseudonymous, and never linked to a ballot, including 0515 0293.

THE COUNT THAT MUST BALANCE, AT EVERY NODE (OBJECTION 18)

Example (hypothetical): one Node's day

962 Vote Coins burned

= 1,000 ballots cast

23 provisional · 15 cast without the card

Spoiled ballots are logged apart (12 in this example): paper that was never cast is never counted.

Burns are public, so anyone can check the balance per Node; provisional and card-less ballots are counted and published separately. A Vote Coin burns once, so no right can appear used twice; check-in stops a second ballot.

The deposit

goes out with the coin and comes back at the burn — or at the close-out burn after certification — never to the voter.

$$N_{\text{cast}} = N_{\text{burn}} + N_{\text{prov}} + N_{\text{no card}}$$

N_{cast} voters' ballots cast at the Node

N_{burn} Vote Coins burned there

N_{prov} provisional ballots

$N_{\text{no card}}$ regular ballots cast without the card

*Burned once, never reissued,
provable by anyone.*

The token lifecycle of Section 14.1 and the balance that follows from Objection 18, as the coin runs in Phases I and II; from Phase III the same right is a spent credential tag in each batch record (Section 12.1). The example numbers are hypothetical.

Check it: 962 + 23 + 15 = 1,000; spoiled ballots (12) stay outside the sum.

- **Close-out.** The burn-only validator permits one more action, and only after the election is certified and its challenge period has closed: a 3-of-3 transaction burns every unused Vote Coin of that election — the coins of registered voters who voted by mail, voted without the card, or did not vote — returns each deposit to the jurisdiction, and publishes the count. Every coin minted is then accounted for: minted equals burned at the polls plus burned at close-out.

14.2 Implementation Requirements

- **Hardware.** Shielded Secure Node enclosures (tested under IEEE Std 299; TEMPEST certification where federally sponsored); five archival-grade industrial printers per Node; high-speed robotic filing systems; write-once media drives and sealed media stock; UPS and solar power systems; and, outside the voting system, anchoring workstations.
- **Software.** Cardano node software on the anchoring workstations only, never inside a Secure Node; the Hyperledger Indentus DID software development kit (SDK); custom **Plutus** (Cardano's smart-contract language) scripts for the minting policy and the burn-only validator; read-only Gold Image OS for voter tablets. All software is published open-source with reproducible builds.
- **Network.** No external connection at any Secure Node; a wired local network isolated inside each Node site. Anchoring workstations use two independent connections from different providers.
- **Staffing.** During certification events and any privileged access, three on-site Triad representatives; during normal voting hours, ordinary trained election workers plus one systems technician — the Triad is an exception-handling and certification body, not a daily staffing requirement.

For the county clerk, the same requirements at a glance, all of them drawn from the sections cited; Phase I exists to put a cost on each line (Objection 37).

One Secure Node	What it takes	Where it is specified
People during voting	Ordinary trained election workers plus one systems technician; the Triad attends only certification before voting, privileged access and exceptions	Section 14.2; Objection 39
Equipment	Shielded enclosure; voter tablets on a read-only Gold Image; five archival-grade printers; one ballot scanner; a filing robot; write-once media drives and sealed media stock; UPS for four hours plus solar	Sections 4.3, 7, 14.2
Capacity	Printing 1,500 ballots an hour (300 per printer); filing 1,000 an hour; Phase I peak 500 an hour	Sections 4.3, 12.3
What leaves the Node	Signed batch records on write-once media, one batch an hour (at least 100 ballots, at most 4,096), carried under two-person custody to the anchoring workstation; per-voter receipts	Sections 4.3, 8.1
Election night	Unofficial totals from the paper; anchors final 36 hours after they land; public dashboard within 72 hours	Sections 8.1, 12.1, 13.2
If something breaks	A failed printer costs one-fifth of capacity; a failed scanner or robot means hand filing and later reconciliation; a multiple failure means hand-marked ballots into the same archive — polls stay open	Section VII; Objection 41
Cost	To be measured in Phase I; a Node-Lite configuration (shared enclosure, three printers, manual filing) is planned for small jurisdictions	Objection 37

PART SIX

Certification & Rights

Sections XV – XVIII · Integrity Invariants · Phased Testing · Voter Bill of Rights · EAC Technology Consultation

WHO THIS PART IS FOR

EAC staff, VSTL engineers, and certification bodies — though Section XVII, the Voter Bill of Rights, is written for every citizen.

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XV. Technical Appendix: Proofs of Integrity

Each invariant below is stated here; its proof is the cited construction (RFC 9162 for inclusion; PLONK for eligibility; Cardano's ledger rules for custody and determinism). The eligibility proof is made by the voter's card at check-in and checked inside the Node.

15.1 Reconciliation Invariant

As formalized in Section VIII, the paper and digital records of a ballot agree when their salted fingerprints are equal ($Vd = Vp$). If $S = 0$, the discrepancy is recorded and resolved from the paper ballot (Section 8.2).

15.2 Zero-Knowledge Proof of Eligibility

The protocol uses a **zero-knowledge proof (ZKP)** — a cryptographic method by which one party proves a statement is true without revealing *why* it is true — in the **zk-SNARK** family (Succinct Non-interactive ARGuments of Knowledge: tiny proofs, checkable in milliseconds). The construction is PLONK-class (Gabizon, Williamson & Ciobotaru, 2019), chosen because its **universal trusted setup** — a one-time, multi-party initialization ceremony that stays sound as long as at least one participant is honest (Bowe, Gabizon & Miers, 2017) — avoids the per-circuit setup burden of older schemes such as Groth16 (Groth, 2016), and aligns with the proof system used by Midnight, the Cardano-affiliated privacy network. Let L be the set of valid eligibility credentials in the Voter Registration Database (VRD):

For every valid credential $c \in L$: $V(x, P(x, w)) = 1$

x = public statement (the published commitment to L for this election)

w = private witness (the voter's credential c and its membership proof — never revealed)

Each voter receives a receipt printed with a tracking code — the ballot's fingerprint, the same value the tablet displays for the voter's approval — and never the validation index. Once the ballot's batch is anchored, the voter, or anyone, can fetch the inclusion proof for that code from the public verification site and check it against the anchored root. Without the sealed salt, no one can match the tracking code to what the ballot says, so the receipt proves nothing to a coercer while the log stays sealed. That condition is stated because it matters: if the log were ever opened, a saved receipt could be matched to its ballot for the rest of the retention period. So the log opens only with all three key-holders present, decrypting it needs key shares held by all three (Section 15.3), and it is destroyed with the paper ballots when the retention period ends. What the ballot says, the voter confirms on the paper itself.

15.3 The Tripartite Authorization Condition

$C \Leftrightarrow \text{Sig}(K(\text{cust})) \wedge \text{Sig}(K(\text{tech})) \wedge \text{Sig}(K(\text{pub}))$

Release of the audit record, C , requires all three oversight signatures. The release is authorized by a 3-of-3 Cardano Native Script transaction, which the ledger rejects unless all three sign, and the sealed record itself opens only with all three key-holders present (Section VI) and is encrypted so that decrypting it requires key shares held by all three — a control that holds even against a copied file. A native script can never be edited, and moving authority to a new one is itself a 3-of-3 transaction — a self-reinforcing governance loop with no bypass.

15.4 Vote Coin Non-Reusability Invariant

$$\forall VC_i : \text{spent}(VC_i) \Rightarrow \neg \exists VC_i'$$

Once a Vote Coin is spent (burned at ballot submission), no identical or replacement token can be issued to the same DID for the same election cycle. The locked Minting Policy Script enforces this, and the EUTXO model's native double-spend prevention guarantees it cryptographically in Phases I–II. From Phase III the same invariant is kept by publication: a credential tag may appear in only one anchored batch list, so a repeated tag is a public, detectable event (Section 12.1), and check-in still prevents a second ballot at the moment of voting.

15.5 EUTXO Determinism Guarantee

Cardano's EUTXO model offers a property that account-based chains such as Ethereum do not: **transaction determinism**. An anchoring or burn transaction can be fully simulated off-chain before submission; if the simulation succeeds, on-chain execution is guaranteed to produce identical results. An anchor therefore lands exactly as built or not at all, and because every signed batch carries a sequence number, a missing anchor shows as a gap instead of disappearing silently.

FIGURE 22 · SECTION XV · PART SIX

Five invariants

The integrity invariants of Section XV, each with what it guarantees

15.1 Reconciliation

$$V_d = V_p$$

each ballot's paper is checked against its record; any mismatch is caught for 0293: 880d738d...08e537d6 on both sides, so $S = 1$ (Figure 11)

15.2 Eligibility, in zero knowledge

$$V(x, P(x, w)) = 1$$

for every valid credential c in L , the proof checks — without revealing which credential, or whose (w is never shown)

15.3 Three-key release

$$C \Leftrightarrow \text{Sig}(K(\text{cust})) \wedge \text{Sig}(K(\text{tech})) \wedge \text{Sig}(K(\text{pub}))$$

the audit record is released only with all three oversight signatures
1 of 8 key combinations (Figure 9)

15.4 No reuse

$$\text{spent}(VC_i) \Rightarrow \neg \exists VC_i'$$

once a Vote Coin is burned, no replacement can exist for that voter in that election (Figure 21)

15.5 Determinism

$$\text{simulate}(\text{tx}) = \text{execute}(\text{tx})$$

an anchor lands exactly as built or not at all; each batch's sequence number makes a missing anchor show as a gap

Each invariant is stated in Section XV (15.5 in words); these are their plain-language readings.

5 invariants

- 15.1 paper checked against record
- 15.2 eligible, but anonymous
- 15.3 three keys (or an alternate), or nothing
- 15.4 one right, one use
- 15.5 as built, or not at all

Five invariants, each one checkable.

Section XV's integrity invariants, each proved by its cited construction. Only 15.1 is about a particular ballot, so only it carries 0293; the others hold for every voter, key-holder and transaction.

Check it: 15.1 is recomputed in Figure 11, 15.3's eight combinations are drawn in Figure 9, and 15.4's single burn is drawn in Figure 21.

XVI. Phased Testing & Certification Schedule

Each phase transition requires Tripartite sign-off on the audit record and completion of every milestone. Phases I and II run as supervised pilots under the host state's authority for non-standard or experimental equipment (for example, Tenn. Code Ann. § 2-9-110). Testing is conducted by an EAC-accredited VSTL engaged by the manufacturer. Each gate must be met before the next phase begins; no phase runs beyond the authority of the one before it. The Phase I gate bounds random reconciliation errors; targeted attacks are tested separately, by the Phase II penetration test and the Phase III red team.

Phase	Name	Scope	Duration	Key Milestone	Certification Gate
I	Alpha	Municipal	90 days	Robotic stress test: 1,000 prints/scans/files per hour sustained for 8 hours	VSTL sign-off + Tripartite sign-off on the audit record; zero unexplained ARP discrepancies across at least 300,000 reconciled ballots (live and robotic test ballots combined), which, treating each reconciliation as an independent trial with the same error rate, shows the rate is below 0.001% at 95% confidence (Hanley & Lippman-Hand, 1983)
II	Beta	County	180 days	Logic validation: ARP mismatch and error-trigger verification across 10,000 simulated votes	Independent penetration test with zero critical findings; full COOP drill completed
III	Gamma	State	1 year	Security hardening: independent air-gap penetration testing and red-team engagement	EAC certification (or the state's equivalent) and state election authority approval
IV	Delta	National	Ongoing	Integration with voter-registration systems; national-scale load testing	EAC federal certification, including the penetration testing federal law now requires (52 U.S.C. § 20971(e))

FIGURE 23 · SECTION XVI · PART SIX

Four phases, four gates

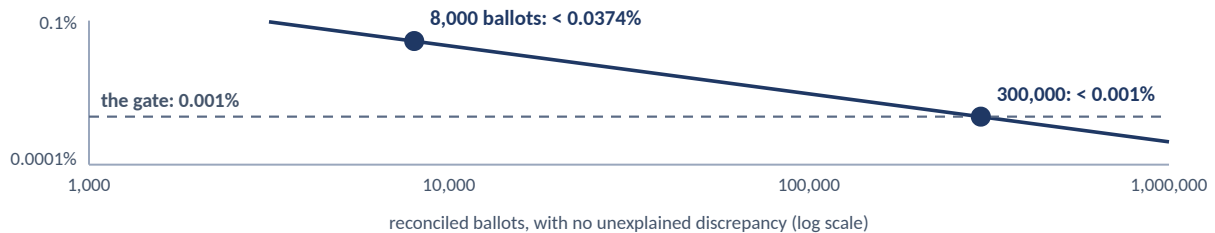
The testing schedule, and the arithmetic behind the Phase I gate

I Alpha • Municipal 90 days MILESTONE Robotic stress test: 1,000 an hour for 8 hours GATE VSTL and Triad sign-off; zero unexplained ARP discrepancies in at least 300,000 ballots	II Beta • County 180 days MILESTONE Logic validation: mismatch detection verified across 10,000 simulated votes GATE Independent penetration test with zero critical findings; full COOP drill	III Gamma • State 1 year MILESTONE Security hardening: air-gap penetration testing and a red team GATE EAC certification, or the state's equivalent, and state approval	IV Delta • National ongoing MILESTONE Voter-registration integration; national- scale load testing GATE EAC federal certification, with the penetration test the law requires
--	--	--	--

Phases I and II run as supervised pilots under the host state's authority; each transition needs Triad sign-off on the audit record.

Why the Phase I gate needs 300,000 ballots

If n independent ballots, all at one error rate, show no unexplained discrepancy, the 95% upper bound on that rate is $1 - 0.05^{1/n}$, about $3/n$ — the “rule of three” (Hanley & Lippman-Hand, 1983).



$$p < 1 - 0.05^{1/n} \approx 3/n$$

$$n \geq \ln 0.05 \div \ln(1 - 0.00001) = 299,572$$

so the gate asks for at least **300,000**

the 8-hour stress test alone (8,000) shows only

$$p < 0.0374\%$$

*A gate is only as strong
as the test that opens it.*

Section XVI's four phases; 300,000 reconciled ballots is 300 hours at 1,000 an hour. Each gate opens the next phase, counts live and test ballots together, and bounds random error only — attacks are tested in Phases II–III.

Check it: $\ln(0.05) \div \ln(0.99999) = 299,571.7$, rounded up to 299,572; and $1 - 0.05^{1/8,000} = 0.0003744 = 0.0374\%$.

XVII. Voter Bill of Rights

Every citizen voting under this protocol is guaranteed the following rights, enforceable through the Tripartite governance structure:

1. **Right to privacy.** Your ballot choice is cryptographically protected. No individual, agency, or authority can link your identity to your vote.
2. **Right to verification.** You receive a receipt that lets you confirm your ballot was included in the public record — and that reveals nothing about how you voted, even to you-under-pressure.
3. **Right to a physical audit.** Your paper ballot is retained and retrievable by independent auditors for the full legal retention period.
4. **Right to correct your ballot.** If the check finds your printed ballot and the digital record disagree before your ballot is filed, you spoil it and mark a new one on the spot. If a disagreement is found later, your paper ballot — the one you read — is what counts.
5. **Right to accessible voting.** Full accommodations — audio guidance, haptic feedback, language assistance — at no additional burden.
6. **Right to transparency.** Aggregate results and audit statistics are published to a public dashboard within 72 hours of poll close.

Vote Coin: one person, one vote — provable by anyone, forgeable by no one.

XVIII. Engaging the EAC: Request for a Technology Consultation

To: Program Director, Testing and Certification, U.S. Election Assistance Commission

CC: National Institute of Standards and Technology; Coordinator of Elections, State of Tennessee (or the pilot state)

Subject: Request for an Early Consultation on New Technology (Program Manual v3.0, §3.3) — The Vote Coin Protocol

Reference Standard: Voluntary Voting System Guidelines (VVSG) 2.0, adopted February 10, 2021; tracked against the draft VVSG 2.1 (September 2025 revision)

18.1 Executive Summary for Regulators

The Vote Coin Protocol is a paper-ballot voting system with a cryptographic audit augmentation. Unlike prior blockchain voting proposals that treat a digital ledger as the ballot, this protocol treats the Cardano blockchain strictly as an audit layer: the voter-verified paper ballot remains the ballot of record, and the system is software-independent in the sense VVSG 2.0 requires — no undetected software error or compromise can change an election outcome, because the paper can always be counted by hand. The blockchain's contribution is to make the audit trail public, per-ballot, continuous, and independently verifiable by any citizen.

The protocol is designed to the VVSG 2.0 requirement set. A complete Technical Data Package will be prepared when a manufacturer is engaged; this paper is offered as the design basis for that work and for an early consultation with EAC staff.

18.2 Consultation Request

I, Michael Twigg, PhD, Principal Architect of the Vote Coin Protocol, request an early consultation with EAC Testing and Certification staff on the new technology this paper describes, under §3.3 of the Program Manual. Any submission of this paper to the EAC will be accompanied by a dated, signed cover letter.

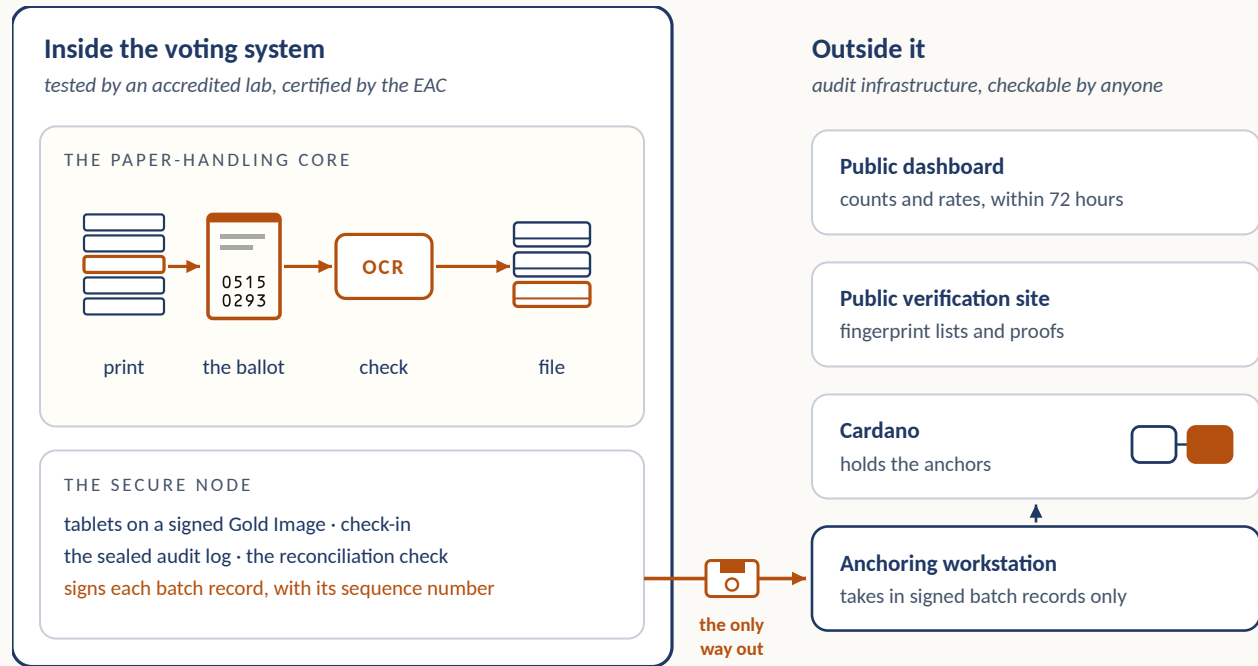
18.3 Next Steps Requested

1. **Consultation.** A meeting with Testing and Certification staff under §3.3 of the Program Manual to review the protocol's new technology — per-ballot reconciliation and public anchoring — and the testable requirements that would apply.
2. **Certification boundary.** Guidance on which components fall inside the voting system (the paper-handling core and the Secure Node) and which sit outside it (the anchoring workstation and public verification tools) — including whether a hardware data diode could carry batch records out (Section 4.3), a question the EAC answers formally through a Request for Interpretation once a manufacturer or test laboratory is engaged (Program Manual v3.0, §9).
3. **Pilot.** Guidance on a supervised Phase I pilot under state authority, with results shared with the EAC before any application.

FIGURE 24 · SECTION XVIII · PART SIX

Where the certification line falls

What the protocol proposes to certify, what stays outside, and what it asks the EAC



WHAT SECTION 18.3 ASKS THE EAC FOR

- 1 A consultation**
on the new technology:
per-ballot reconciliation
and public anchoring (§3.3)
- 2 Guidance on this line**
which components fall
inside the voting system,
and which outside it
- 3 Guidance on a pilot**
a supervised Phase I
under state authority,
results shared first

The line drawn here is the paper's proposal: the EAC decides where it falls.

1 line, 1 way out

inside: the paper-handling core and the Secure Node
outside: the anchoring workstation and the public verification tools

*Certify the paper system.
Keep public anchoring outside it.*

Section XVIII's request for a technology consultation. Nothing outside the line can change a ballot: everything that crosses out is a batch record already signed inside the Node.

Check it: against Section 18.3, item 2 — “the paper-handling core and the Secure Node” inside; “the anchoring workstation and public verification tools” outside.

PART SEVEN

Objections & Rebuttals

Section XIX · Fifty Questions, Answered

WHO THIS PART IS FOR

Skeptics first — this Part exists for them. Also journalists, legislators, staffers briefing decision-makers, and anyone who arrived at this document already convinced it must be a bad idea.

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XIX. Anticipated Objections & Rebuttals: Fifty Questions, Answered

A proposal that touches elections will be attacked from every direction, and it should be. This section collects the fifty strongest objections the architect expects — technical, legal, political, economic, and human — and answers each directly. Readers are encouraged to start with whichever objection they find most damning.

A. Security & Engineering (1–13)

1. “Peer-reviewed research says blockchain voting is a bad idea. Full stop.”

That research — most prominently Park, Specter, Narula, and Rivest (2021) — targets *internet* voting with a blockchain as the ballot box. This protocol agrees with their conclusions: it forbids internet voting entirely, keeps voting in person, and makes the paper ballot the ballot of record. The chain is an audit annex, not the election. The cited critique is an argument *for* this architecture, not against it.

2. “If the voter tablet is compromised, everything downstream is garbage.”

The tablet is treated as untrusted by design. It boots from a signed, read-only image, is wiped between voters, and — decisively — the vote it produces is not the vote. The voter reads their selections in plain text on the printed ballot before casting it. A tablet that prints something other than what the voter chose is caught by the voter, spoiled, and reprinted; a tablet that prints correctly but records something else is caught when the ARP compares the scanned text of the paper with the digital record.

3. “This violates software independence — VVSG 2.0’s core principle.”

It satisfies it in the strictest sense. Software independence means no undetected software change can cause an undetectable change in the outcome (Rivest, 2008). Here the outcome is determined by paper ballots that can be hand-counted at any time, and every digital component’s output is checked against that paper, ballot by ballot. The software could be deleted after the election and the result would still be provable. Its one dependency is the voter’s own check of the paper, which Objection 7 addresses.

4. “Air gaps fail. Stuxnet crossed one.”

Stuxnet crossed an air gap via removable media carried by insiders (Falliere, Murchu & Chien, 2011) — which is why this design never treats the air gap as the whole defense. The Secure Node has no network connection in or out, it is electromagnetically shielded, and the Three-Key Rule requires three independent institutions to be physically present for any access to its cryptographic or storage hardware. The two data paths that must exist run as ceremonies on write-once media: the election definition comes in before polls open — signed, checked against its published fingerprint, and loaded only with all three keys present — and signed batch records go out for anchoring on fresh media from sealed stock, under two-person custody. And if all of that failed anyway, the attack still faces the paper: a full manual recount remains available and legally sufficient.

5. “Executive Order 14248 moves to ban barcodes and QR codes on ballots. Doesn’t that sink this design?”

It does the opposite — the protocol was built for exactly this. Executive Order 14248 (2025) directed the EAC to bar ballots in which a vote is contained within a barcode or QR code, except where needed for voters with disabilities. A federal court has since held that directive unlawful as a violation of the

separation of powers and enjoined it (*Washington v. Trump*, W.D. Wash. 2026). The government appealed in March 2026, and in May 2026 the Ninth Circuit paused the appeal until the district court finishes the case, so anyone relying on this ruling should check its current status. Either way, the EAC's own draft VVSG 2.1 carries the same rule, and it is the right rule. This protocol contains no vote in any code. The ballot carries the voter's selections in plain, human-readable text, and its only machine-readable element is a validation index — a lookup number set in the OCR-B typeface standardized in ISO 1073-2 and read by optical character recognition — the same font and technology used to read the machine-readable zone of every passport issued under the international ICAO standard (ICAO Doc 9303). The scanner reads the identical characters the voter reads; there is no separate encoded layer, so there is nothing for the order to prohibit.

This also dissolves the failure that made machine codes suspect in the first place. In the 2023 Northampton County, Pennsylvania incident (Votebeat, 2025), a machine code and the printed text disagreed. Here they cannot disagree, because there is only one representation: the human-readable text is what the voter verifies, what the scanner reads, and what a hand count counts. A rare OCR misread costs a second read and, at worst, a re-marked ballot (Section 8.2), never a vote — the paper the voter read remains the record.

The design borrows a proven precedent. Casino Ticket-In / Ticket-Out systems, governed for two decades under the GLI-13 validation standard and written into state regulation (for example, N.M. Admin. Code § 15.1.7.11), print a validation number that carries no value itself; the money lives in a protected central record the number merely points to, reconciled in real time before any payout. The gaming precedent is not an “escape the ban” move — it is proof that the index-not-vote architecture is battle-tested in an adversarial, high-stakes environment. What the precedent proves is the pattern — a printed number that carries no value and points to a protected record; ballot secrecy and the paper record are not gaming questions, and the design meets them on its own terms (Sections VIII and IX). Vote Coin applies the same discipline to the ballot: the code is a pointer, the paper is the truth. This is the fullest expression of the federal vision — a voter-verifiable paper record counted from what the voter can read — and it meets every state's secret-ballot and paper-record requirement without asking anyone to trust a code they cannot read.

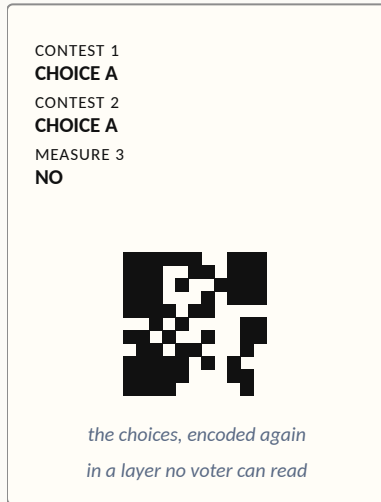
Nor is this untested in elections. The first voting system the EAC certified to VVSG 2.0 prints human-readable summary ballots with no barcode or QR code and counts them by optical character recognition (EAC, 2025b) — and the lab that tested it belongs to the same organization that publishes the GLI-13 casino standard.

FIGURE 25 · OBJECTION 5 · PART SEVEN

One representation

Why a ballot with no encoded layer has nothing for a barcode ban to touch

A BALLOT WITH A MACHINE CODE

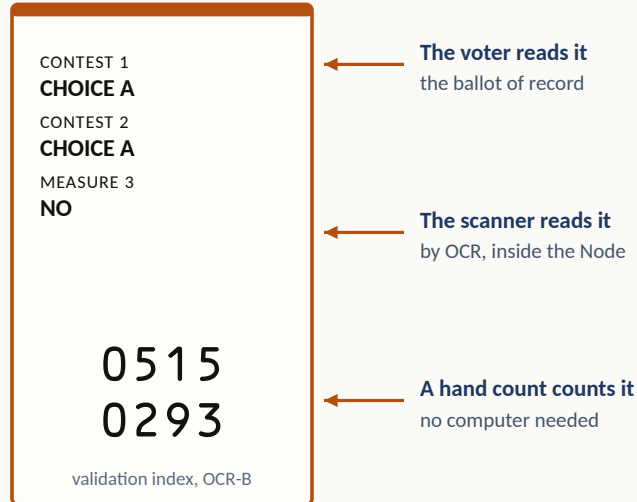


Two representations of one vote.

If they disagree, which one counts?

In Northampton County, Pennsylvania, in 2023, a machine code and the printed text disagreed.

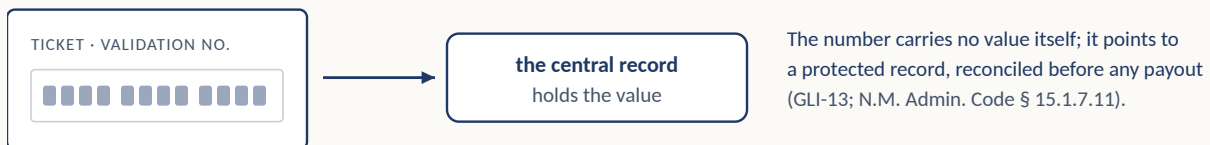
BALLOT 0515 0293 — NOTHING ENCODED



One representation. The voter and scanner

read the same printed characters, so no second code can contradict them. A rare OCR misread costs at worst a re-marked ballot, never a vote.

THE PRECEDENT: A CASINO TICKET



The gaming precedent is not an “escape the ban” move — it is proof that the index-not-vote architecture is battle-tested.

The drawn code at left is generic, not a real symbol; the casino ticket is shown only for its validation number.

1 representation

what the voter reads = what the scanner reads
= what a hand count counts
no barcode, no QR code, no encoded layer

*The index is a pointer.
The paper is the truth.*

Objection 5. EO 14248 directed the EAC to bar votes held in a barcode or QR code (a court has since enjoined that directive); the draft VVSG 2.1 carries the same rule. Here the choices are plain text, and the only other machine-read element is a human-readable index, which the draft expressly permits.

Check it: the first line of 0293's record, INDEX: 05150293, is rebuilt from the printed 0515 0293 by removing the space (Figure 11).

6. “OCR tabulation is insecure — Princeton’s Andrew Appel said so in 2026.”

He was describing a specific plan: ballot images uploaded to a vendor’s servers and read there (Appel, 2026). His concerns — images altered in weakly secured databases, anyone at the processing company changing inputs or outputs, a scanner deciding what is recorded — are real. Here, OCR runs inside the air-gapped Node and no image leaves it. The OCR result decides nothing on its own: it only checks the paper against the digital record, and any disagreement is settled by people reading the paper. The paper remains the ballot of record, audited by hand. What remains of his critique — whether voters check printed ballots — is answered in the objection that follows.

7. “Voters don’t check machine-printed ballots. A hacked tablet that prints and records the same wrong choice beats your whole design.”

This is the strongest critique of any ballot-marking design (Appel, DeMillo & Stark, 2020), and the research supports it: in one realistic mock election, without intervention, only 40% of participants reviewed their printed ballots at all and only 6.6% told a poll worker about the planted error (Bernhard et al., 2020). The reconciliation check cannot catch a tablet that prints and records the same wrong choice — only the voter can. So the protocol does four things. Every voter may hand-mark a pre-printed paper ballot instead of using the tablet; it goes into the same archive, its scanned record is fingerprinted and anchored like any other, and it is covered by the same audits. The tablet requires an explicit review step before printing, and poll workers ask every voter to read the paper — prompts that research shows raise review and reporting (Bernhard et al., 2020; Kortum, Byrne & Whitmore, 2021). Spoiled-ballot rates are tracked per device in real time, so a tablet that voters keep correcting is pulled. And devices are spot-checked during voting with test ballots cast by the Triad’s citizen-oversight member — printed with a TEST mark, filed in a separate box, excluded from the tally and from the burn balance, and counted in the public log. None of this makes a ballot-marking device perfect; it makes a silent attack much harder to keep silent.

FIGURE 26 · OBJECTION 7 · PART SEVEN

If a tablet lies

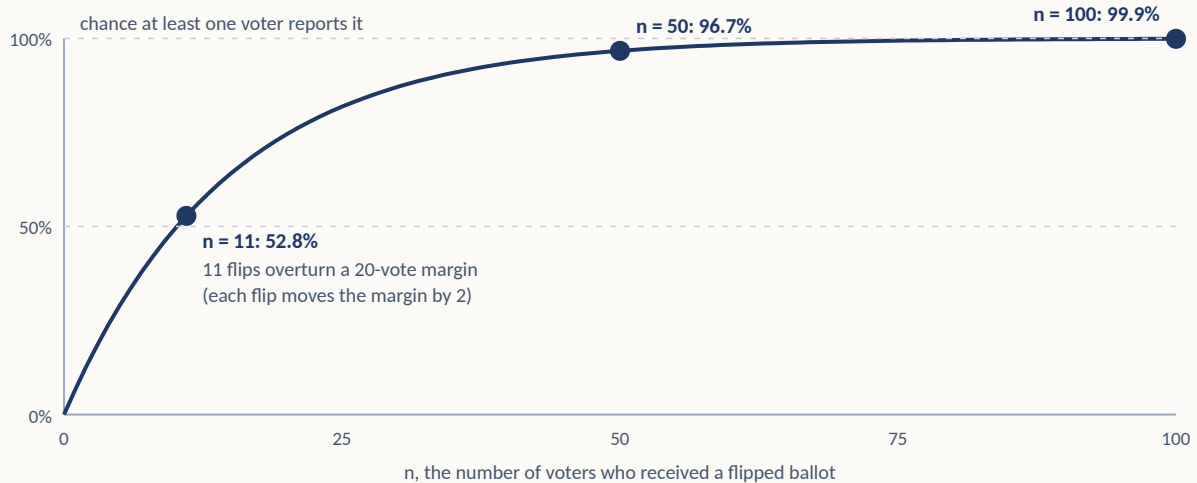
What it takes to catch a tablet that prints and records the same wrong choice

THE ATTACK THE RECONCILIATION CHECK CANNOT SEE (A WHAT-IF)



A hacked tablet prints and records the same wrong choice. Paper and record agree, so the check passes. Only a reader of the paper can notice.

In one mock election, without prompts, 40% of participants reviewed their ballots; 6.6% told a poll worker of the planted error (Bernhard et al., 2020).



SO OBJECTION 7 ADDS FOUR SAFEGUARDS

- | | | | |
|--|--|--|---|
| <p>1 Hand-marked paper
any voter may skip the tablet; same archive, same audits</p> | <p>2 Review, then read
an explicit review step, and poll workers ask every voter to read it</p> | <p>3 Spoiled-ballot rates
tracked per device, live; a tablet voters keep correcting is pulled</p> | <p>4 Spot checks
test ballots cast during voting by the citizen-oversight member</p> |
|--|--|--|---|

Prompts like safeguard 2 raise review and reporting (Bernhard et al., 2020; Kortum, Byrne & Whitmore, 2021). The chart uses only the no-prompt 6.6%; no number is claimed for any safeguard until a source supports it.

None of this makes a ballot-marking device perfect; it makes a silent attack much harder to keep silent.

$$P = 1 - (1 - d)^n$$

$d = 0.066$: the study's rate, without prompts

n voters given a flipped ballot

P chance at least one reports it, if voters act independently and one report is acted on

*On its own, a 6.6% report rate is weak.
That is why there are four safeguards.*

Objection 7, with the arithmetic reviewers will do themselves. The curve uses only the study's no-prompt 6.6%; it is the baseline before the safeguards, not an estimate of them.

Check it: $1 - 0.934^{11} = 0.528$; $1 - 0.934^{50} = 0.967$; $1 - 0.934^{100} = 0.9989$.

8. “A malicious printer could print one thing and record another.”

The printed text is verified by the voter before deposit, and the deposited ballot is independently re-scanned by a second machine that reads its printed selections and index; their salted fingerprint must match the digital record’s (Section VIII). Fooling this requires the printer, the independent scanner, the ledger, and the voter’s own eyes to fail in perfect coordination — and even then, sampled hand audits of the paper remain.

9. “Hash collisions could let two different ballots share one fingerprint.”

SHA-256 offers 128-bit collision resistance, NIST’s rating: finding a collision is expected to take on the order of 2^{128} — about 3.4×10^{38} — hash computations. If that assumption ever weakened, the hash function is a replaceable module — and the paper ballots remain the record regardless.

10. “zk-SNARKs require a trusted setup. Whoever ran it can forge proofs.”

The protocol specifies a PLONK-class proof system with a universal, updatable setup: a one-time public ceremony that remains sound as long as *any single participant* among many was honest (Bowe, Gabizon & Miers, 2017), and that can be re-run or extended at will. Transparent (setup-free) proof systems are a viable fallback if certification bodies prefer them, and being hash-based, they are also a post-quantum path (Objection 20).

11. “Electromagnetic side channels could leak votes from inside the node.”

Shielding against this threat is a mature, decades-old engineering discipline; the U.S. government’s TEMPEST program is its best-known form (National Security Agency, 2007). Enclosures are shielded to a published standard and tested under IEEE Std 299, RF-emitting devices are excluded from the floor, and the Triad’s technical-certification member verifies the shielding before polls open — never during voting.

12. “Hardware can be backdoored at the factory, before you ever see it.”

True of every voting system ever built — which is why this one dual-sources critical components, requires tripartite acceptance testing, publishes its hardware bill of materials, mandates reproducible software builds, and specifies third-party hardware attestation before Phase I. And beneath all of it: the paper record means a factory backdoor can corrupt convenience, not outcomes.

13. “Attackers will DDoS Cardano on Election Day.”

Let them. A distributed denial-of-service attack floods a network with traffic to slow or stop it, and the chain is not in the critical path: casting is local-first, and anchoring is asynchronous. A congested or even fully unavailable network delays public anchoring by hours, not a single vote by a second. Cardano has already been tested this way. On November 21, 2025, a single malformed transaction exploited a node bug and split the chain until patched nodes rejoined it (Intersect, 2025). An election running that day would have lost nothing: ballots are recorded locally, and an anchor counts as final only after the chain has settled past it — 129,600 slots, 36 hours, as published with the batch rules (Section 8.1). The only effect would have been a later public anchor.

B. Blockchain-Specific (14–21)**14. “You don’t need a blockchain. A signed database does this.”**

A signed database proves the database operator signed something; it cannot prove to an outside skeptic that records weren’t rewritten before signing, backdated, or selectively deleted — the auditor must trust the operator. A public ledger removes that trust requirement: anchors are visible to

everyone, ordered by a consensus no single party controls, and checkable forever. The blockchain's job is narrow but irreplaceable: making the audit trail *nobody's* property.

15. “Why Cardano? This looks like vendor lock-in to one chain.”

Cardano was chosen on engineering merits: EUTXO determinism (transactions cannot half-succeed), native assets (tokens tracked by the ledger itself, with one small, audited validator as the only custody code), a formal-methods research culture, and negligible energy use. But the architecture touches the chain through a narrow anchoring interface, so anchors can be re-published to another ledger without touching a single ballot.

16. “Cardano isn’t American. Foreign infrastructure shouldn’t touch our elections.”

Cardano is open-source software run by anyone, anywhere — including the sovereign, domestically operated nodes this protocol specifies. No foreign entity operates, approves, or can interfere with a Secure Node. The decentralization is the point: no country, company, or foundation — American or otherwise — can rewrite the ledger.

17. “A stake-majority attack could rewrite the chain and the election with it.”

Rewriting Cardano's history is an attack costing billions and visible to the entire world in real time — and it still couldn't change a single vote, because votes live on paper and in local cryptographic logs. Anchors would simply be republished (optionally to multiple chains). The attacker buys global notoriety and changes nothing.

18. “The token adds nothing. It’s a database row wearing a costume.”

The token converts “the administrator promises each voter voted once” into a public record anyone can check. A Vote Coin can be burned only once, so no voting right can appear as used twice, and each Node's burns — from Phase III, the credential tags in its batch records (Section 12.1) — must equal its ballots cast by voters who used their card; provisional ballots and ballots cast without the card are counted and published separately, and spoiled ballots are logged apart. Check-in still stops a second ballot at the moment of voting, as it does today; the token proves afterward that it did, in a record no operator holds and root access cannot rewrite. Removing trust in the operator is the entire project.

19. “Calling it a ‘coin’ invites speculation and gift.”

There is nothing to speculate on: the token sits at a burn-only validator address and can never be moved, so no exchange can list it, no market can price it, and no one can buy it — at any price. “Coin” is a naming convention for a native asset, not an economic property. Public-facing materials lead with the paper ballot, not the token (Objection 47).

20. “Blockchains are forever. Ballot secrecy has to survive future computers, including quantum ones.”

Nothing secret is on the chain — no ballot content, no identity, only salted hashes and pseudonymous token events. A future adversary with a quantum computer and the full public ledger learns that votes occurred and were reconciled; it cannot learn how anyone voted, because no ballot content is ever written to the chain: its fingerprints are salted with 256-bit secrets that never leave the sealed audit log. Signatures are the part that must change. NIST's post-quantum signature standards (FIPS 204 and 205) are the migration path for signing batch roots, on NIST's draft timeline to retire elliptic-curve signatures by 2035 (NIST IR 8547); hash-based fingerprints are expected to remain sound, and a transparent, hash-based proof system is the fallback for the eligibility proof.

21. “Your costs depend on ADA’s price. Crypto volatility funds your election?”

Anchoring a batch root is cheap: one small transaction covers thousands of ballots, and at September 2026 settings even the largest transaction Cardano allows costs under 1 ada in fees (0.155 ada plus 0.000044 ada a byte; Section 12.1). Per-voter burns are a different load, which is why they stay on the main chain only through Phase II (Section 12.1). Per-voter token deposits are larger (Section 4.2) but refundable — the jurisdiction recovers them at the burn, or at the close-out burn after certification (Section 14.1). Both come from the jurisdiction’s appropriated election budget, not from market speculation. There is no treasury and no staking income: these are line items in an ordinary election budget, paid in ada when used, and small enough that price swings cannot threaten an election.

C. Privacy & Civil Liberties (22–28)**22. “Facial recognition at the polls is dystopian and biased against minorities — and a biometric database breach would be catastrophic and irreversible.”**

Face matching is optional. A voter who opts in stores a face template only on their own credential card; at the polls it is matched on the device against that card, and nothing is retained after the session. There is no central face database, so there is no honeypot to breach. Algorithms are restricted to those benchmarked in NIST’s Face Recognition Technology Evaluation (FRTE, formerly FRVT), with a false-match rate of 1 in 10,000 or better for every demographic group and presentation-attack detection, as NIST SP 800-63B-4 requires. A voter who declines uses the card and PIN alone, and the ordinary poll-book check-in remains available to everyone, exactly as today.

23. “Any receipt enables coercion: ‘show me your receipt or else.’”

The receipt is engineered to be useless to a coercer. It carries a tracking code whose inclusion proof shows that the ballot’s fingerprint was anchored, and without the sealed salt no one can match that code to what the ballot says, so it reveals nothing about content. A voter under duress can show it freely — it testifies to participation and nothing else. This holds while the sealed log stays sealed: it opens only with all three key-holders present, is encrypted to their three key shares, and is destroyed with the paper ballots when the retention period ends (Section 15.2). End-to-end-verifiable systems, from the open-audit Helios (Adida, 2008) to ElectionGuard in real U.S. elections (Microsoft Democracy Forward et al., 2023), give voters receipts of this kind; coercion resistance with individual verifiability is the property the cryptographic literature has demanded since Juels, Catalano, and Jakobsson (2005).

24. “An employer could demand proof you voted at all.”

Whether someone voted is already public record in most U.S. states today; the protocol neither adds to nor subtracts from that. What it adds is that proof of *how* you voted cannot exist, for anyone, ever.

25. “The government will secretly link identities to ballots. They always find a way.”

Unlinkability here is not a policy promise — it is a mathematical property of the zero-knowledge construction, reinforced physically by anonymizing ballots before scanning. Even the three Triad institutions colluding after the fact lack the information to reconstruct the link, because it is never recorded anywhere. You cannot leak what was never stored, and the records federal law requires the Node to keep — which the Attorney General may demand (52 U.S.C. §§ 20701, 20703) — contain nothing that connects a voter to a vote.

26. “Today it’s voting. Tomorrow this ID system tracks everything. Mission creep is inevitable.”

The DID architecture is voter-held by design: the citizen controls the credential, and statute limits its electoral use to authentication at ballot issuance. Because the credential lives with the citizen rather than in an agency’s database, the usual instrument of mission creep — quietly repurposing centralized records — has nothing to grab. Sunset and destruction schedules are written into the enrollment consent.

27. “Enrollment burdens will fall hardest on poor and rural communities.”

Enrollment is free, happens once (it persists across every future election), and travels: mobile units are a funded, scheduled program — not an afterthought — covering underserved neighborhoods, care facilities, tribal lands, and correctional institutions. And on Election Day, HAVA’s provisional-ballot safety net remains fully intact.

28. “My grandmother will never vote on a tablet.”

Your grandmother will use a touchscreen for under a minute, with a human official beside her if she wants one, and then hold an ordinary paper ballot in her hands — which is the part that counts. The experience is closer to an ATM followed by a familiar paper ballot than to anything resembling cryptocurrency. She needs no phone and no email address — a card and a PIN, as at a bank machine — and if she prefers, she can skip the touchscreen and mark a paper ballot by hand (Objection 7). Everything cryptographic happens invisibly, on her behalf.

D. Legal & Constitutional (29–36)**29. “Elections are run by states. The Secret Service and the military have no business here.”**

Correct — and the protocol is built to honor it. The Triad is a governance *pattern*: three keys, three independent institutions, zero unilateral action. Section 6.2 specifies jurisdiction-appropriate instantiations — state election authorities, state-controlled National Guard technical units under Title 32, and statutory citizen oversight boards — and the reference model itself is civilian. Where law permits federal agencies to take part, they certify equipment before voting and are never present at the polling place.

30. “Military involvement in domestic elections violates the Posse Comitatus Act.”

The reference model has no military role. Where a jurisdiction chooses a National Guard technical unit, its role is technical certification — shielding and air-gap verification before voting — not law enforcement, not administration, and not presence at the ballot, and in state-controlled status (Title 32) the Posse Comitatus Act does not apply. Separately, federal law makes it a crime to bring troops or armed men to any place where an election is held, or for members of the armed forces to interfere with an election (18 U.S.C. §§ 592–593). That is why every technical-certification task happens before polls open, away from the polling place, and no uniformed or armed personnel are present during voting.

31. “State constitutions mandate a secret ballot. Receipts and ledgers break that.”

The protocol’s secrecy protections match or exceed current practice: the ballot choice is kept only on anonymized paper and in the Node’s encrypted vote packages, never published, the receipt reveals nothing about content, the ledger holds only content-free batch roots, and the validation index is random and never linked to the voter (Section 8.1). Any jurisdiction’s secrecy clause is satisfied.

32. “VMSG 2.0 has no category for this. You can’t certify what has no box to check.”

The protocol’s core is what the EAC certifies routinely: a paper-based voting system — in 2025 the EAC certified one that prints human-readable summary ballots and counts them by OCR (EAC, 2025b). What is new is the audit layer, and the EAC’s Program Manual has a process for exactly that: an early consultation on new technology before any application (§3.3). The blockchain components are audit infrastructure, comparable to other electronic audit tooling, and the consultation requested in Section 18.3 exists to settle these questions before any application.

33. “Federal law requires 22-month ballot retention; blockchains retain forever. Those conflict.”

No conflict exists because no ballot is on the chain. Physical ballots follow statutory retention and destruction schedules exactly as today. What persists on-chain are cryptographic hashes — audit metadata with no ballot content — which retention law does not restrict.

34. “Mandatory in-person voting disenfranchises absentee voters, overseas military, and voters with disabilities.”

The protocol governs the in-person channel only. Absentee, mail, and UOCAVA (overseas and military) voting continue under existing law, unchanged. This version deliberately declines to “solve” remote voting, because the security literature says it cannot yet be solved responsibly; hardening the in-person channel first is the honest sequence.

35. “If enrollment or authentication fails on Election Day, a citizen loses their vote.”

No. HAVA §302’s provisional-ballot requirement is preserved intact: any voter whose credential fails for any reason uses the ordinary poll-book check-in, exactly as today, and any voter whose eligibility cannot be confirmed casts a provisional paper ballot on the spot, adjudicated under existing state procedures. Technology failure can never subtract a voter.

36. “What is the legal standing of a burned token? Courts don’t recognize ‘crypto ceremonies.’”

None is claimed. The paper ballot is the legal instrument, admissible exactly as ballots are today. The token burn is an administrative bookkeeping event — evidence that may corroborate, but never replaces, the ballot. No statute needs amending for the system to operate.

E. Operational & Economic (37–44)**37. “TEMPEST rooms, robots, five printers — the cost per node is absurd.”**

Compare lifecycle to lifecycle, not sticker to sticker: current jurisdictions pay for aging DRE fleets, per-election vendor service contracts, audit litigation, and recount chaos. A Secure Node is durable multi-cycle capital serving many elections, and a scaled-down “Node-Lite” configuration (shared shielded enclosure, three printers, manual filing) is planned for small jurisdictions. Phase I exists to produce real cost data before anyone commits at scale.

38. “Lines. This will create hours-long lines.”

Casting is local and takes minutes; nothing waits on a network. Node capacity is sized to precinct population like any polling place, queue telemetry is monitored in real time, and a mobile reserve fleet deploys where waits grow. The slowest physical step — filing, at 1,000 ballots/hour per Node — runs at twice the Phase I peak, and printing (1,500/hour) at three times it.

39. “Staffing three institutions at every polling place is fantasy.”

The Triad staffs *exceptions*, not operations. Normal voting hours run on ordinary election workers plus one technician. Three-key presence is required only for privileged access: equipment certification before voting, work on cryptographic or storage hardware, archive access, and release of the audit record — routine mechanical maintenance is the technician’s job, in view of two poll workers (Section 4.3) — rare, scheduled events, not daily coverage.

40. “Thermal paper fades. Your ‘permanent’ ballots will be blank in a year.”

The specification requires archival-grade printing and ballot stock rated beyond the 22-month federal retention window, verified by accelerated-aging tests in Phase I. This objection is right about thermal paper, which is why the specification rules it out for a ballot of record.

41. “Printers jam, robots fail, ink runs out. Machines break on the worst possible day.”

That is why five printers run active-active: any single failure costs 20% capacity and zero downtime. Consumables are stocked on site and components are dual-sourced. The filing robot is a convenience, not a dependency: if it fails, ballots are filed by hand in numbered sleeves and scanned with handheld units, so retrieval slows but no vote and no audit capability is lost. And if the improbable multi-failure occurs, the degraded mode is the oldest technology in democracy — hand-marked paper ballots into the same archive, with polls staying open (Section VII).

42. “Rural areas don’t have the connectivity for any of this.”

They don’t need it. A Secure Node never connects to any network: it runs at full capability all day, and its signed batch records travel on write-once media to an anchoring workstation that can sit at the county office. Connectivity affects when the public audit anchor appears, never whether voting works.

43. “Whoever builds and maintains this becomes the new Diebold — vendor capture with extra steps.”

Every design decision cuts against capture: fully open-source software, reproducible builds (anyone can verify the published binary matches the published code), a public Technical Data Package, published hardware bills of materials, and multiple licensed manufacturers. The protocol is a specification anyone may implement — the opposite of a proprietary moat.

44. “The environmental footprint of blockchain is indefensible.”

That criticism belongs to Proof-of-Work chains. Cardano’s Proof-of-Stake consensus, Ouroboros (Kiayias et al., 2017), uses over 99% less energy (Platt et al., 2021), the protocol’s batched anchoring adds a negligible transaction load, and each Node’s solar array offsets its own draw. The system likely retires more energy-hungry legacy hardware than it introduces.

F. Political & Public Trust (45–50)**45. “Proposing this legitimizes conspiracy theories about current elections.”**

Aviation didn’t insult pilots by inventing the black box, and medicine didn’t indict surgeons by adopting checklists. Engineering deeper verifiability is not an accusation — it is what serious institutions do with critical systems. The protocol’s premise is that public confidence should rest on proof rather than reassurance, whichever direction the doubts blow from.

46. “Partisans will weaponize the complexity: ‘you can’t trust what you can’t understand.’”

The system’s answer to every dispute is radical and simple: *count the paper*. No citizen is ever asked to trust mathematics they cannot follow — they can watch a hand recount of human-readable ballots, precinct by precinct. The cryptography exists for those who want more assurance, never as a prerequisite for anyone’s confidence.

47. “The public hears ‘crypto’ and thinks scams. The branding is poison.”

Public-facing materials lead with what the voter experiences: a paper ballot you read and cast, receipted and auditable. The token is invisible plumbing — no voter ever handles a cryptocurrency wallet, a private key, or the word blockchain — only a card and a paper ballot. The technical name matters to evaluators reading this document; the ballot matters to everyone else.

48. “Complexity is itself a vulnerability. Simple systems fail simply; this fails in ways nobody will foresee.”

The complexity is deliberately confined to the *audit* layers, each of which fails safe to the layer below — and the bottom layer is paper counted by hand, the simplest voting system that exists. The voter-facing path is a touchscreen and a printed ballot: simpler than many DREs deployed today. Complexity you can amputate without losing the election is redundancy, not fragility.

49. “Why should anyone trust the architect? Every con starts with ‘trust me.’”

Nothing in this system requires trusting its architect, and that is its central feature. The architect holds no keys, no privileged access, and no proprietary secret: the specification is open, the code is reproducible, governance is 3-of-3 across independent institutions, and any qualified party can implement, audit, or fork the entire design. A system whose author must be trusted has failed this document’s first premise.

50. “Optical scan plus risk-limiting audits already works. This is a solution looking for a problem.”

Optical scan with RLAs is the best system currently deployed, and this protocol is best understood as its upgrade rather than its rival. The additions are concrete: reconciliation of every tablet-printed ballot in real time instead of a sample weeks later; public anchoring at short, regular intervals instead of trust in custody logs; instant retrieval of any physical ballot from its anchored reference; and three-institution custody instead of single-official control. Where an RLA says “the outcome is probably right,” this system says “here is the proof, per ballot, and check it yourself.”

PART EIGHT

Appendices

Section XX · About the Architect · Glossary · References & Standards

WHO THIS PART IS FOR

Reference use — return here whenever a term or citation needs checking.

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XX. About the Architect

Michael Twigg, PhD, is doctoral faculty and a dissertation chair in the College of Doctoral Studies at the University of Phoenix, where he teaches graduate courses in curriculum design, educational technology, instructional media and leadership, and guides doctoral candidates through their dissertation research. His technology career reaches back to 1996 and the founding of OkayNext. The Vote Coin Protocol is the synthesis of pedagogical systems thinking with applied cryptographic engineering, grounded in one conviction: the most consequential democratic infrastructure must ultimately be understood, checked, and trusted by the citizens it serves — not merely operated on their behalf. The views expressed here are his own and do not represent any institution with which he is affiliated.

Glossary

ada — Cardano’s native currency (ticker ADA). Network fees and token deposits are paid in ada, from the jurisdiction’s election budget.

Air gap — Physical isolation of a system from all external networks and devices. Data crosses only on physical media, under controlled procedures.

Anchoring workstation — A computer outside the certified voting system that takes a Secure Node’s signed batch record from write-once media and publishes its root, with the batch’s token burns, in Cardano transactions. It can delay an anchor but cannot alter one.

ARP (Automated Reconciliation Protocol) — The per-ballot check that the salted fingerprint of the digital record matches the salted fingerprint of the scanned paper ballot (Section VIII).

Blake2b-256 — The hash function Cardano uses internally for its own records. The protocol’s ballot checks do not depend on it (see *SHA-256*).

BMD (Ballot-Marking Device) — A touchscreen that prints a paper ballot for the voter to check before it is counted. This protocol’s voter tablets work this way.

Burning (a token) — Permanently destroying a token on-chain, leaving a public, timestamped record that it was used exactly once.

Canvassing authority — The official body that certifies election results under state law.

COOP (Continuity of Operations) — Planning that keeps essential functions running through disruptions. Here the Node’s degraded mode keeps polls open (Section VII).

Credential card — A smart card issued at enrollment that holds the voter’s decentralized identifier and, if the voter chose, a face template. It is unlocked with a PIN at the polls and persists across elections.

Credential tag — The pseudonymous identifier a voter’s right carries: the tag on the Vote Coin in Phases I–II, and the value each batch record lists as spent from Phase III (Sections 12.1, 14.1). Linkable to a voter only by the issuing jurisdiction.

Data diode — A hardware device that physically permits data to flow in one direction only — a one-way valve for information. Proposed here only as a possible substitute for carried media, if an EAC Request for Interpretation finds it within the boundary VVSG 2.0 Requirement 14.2-E permits.

DID (Decentralized Identifier) — A digital identity credential held and controlled by the individual rather than stored in a central database.

DRE (Direct-Recording Electronic) — A touchscreen voting machine that records votes only in electronic memory, historically without a paper trail.

E2E-V (End-to-End Verifiable) — A voting system in which each voter can confirm their ballot was included and anyone can check the published tally, without any vote being revealed.

EAC — U.S. Election Assistance Commission — the federal body that accredits voting-system test laboratories and certifies voting systems.

EUTXO (Extended Unspent Transaction Output) — Cardano’s accounting model, in which a transaction’s outcome can be fully computed before submission, guaranteeing deterministic execution.

FIPS 140-3 — The current federal standard for validating cryptographic modules (successor to FIPS 140-2).

FRTE (Face Recognition Technology Evaluation) — NIST’s benchmark program for face-matching algorithms, formerly called FRVT.

Gold Image — A cryptographically signed, read-only master copy of an operating system, reloaded fresh for every voter session.

HAVA — Help America Vote Act of 2002 — federal law setting core voting-equipment, provisional-ballot, and accessibility requirements.

Hydra — Cardano’s Layer-2 scaling framework: fast local transaction channels that settle their final state to the main chain.

Hyperledger Indentus — The decentralized identity framework used for voter credentialing (formerly known as Atala PRISM).

Merkle tree / inclusion proof — A structure condensing many records into one fingerprint (the root) while allowing any single record to prove it belongs — the mechanism behind batched anchoring (built to IETF RFC 9162; Section 8.1).

MFA (Multi-Factor Authentication) — Proving identity through several independent means (something you are, know, and have) rather than any single one.

Minting Policy Script — The on-chain rule that governs when a Cardano native asset may be created or destroyed.

Multi-signature (multi-sig) — A wallet or policy that refuses to act unless multiple independent keyholders sign together.

Native asset — A token type Cardano’s ledger tracks and enforces directly, alongside ada, without a separate token contract.

OCR (Optical Character Recognition) — Software that reads printed characters. Here it reads the ballot’s printed selections and its validation index, set in the OCR-B typeface.

Oracle problem — A blockchain’s inability to verify real-world facts without trusting an outside reporter.

PIA (Privacy Impact Assessment) — The structured privacy analysis federal law requires for systems that handle personal information (Section IX).

PLONK — A zk-SNARK proof system with a universal, updatable trusted setup usable across all circuits.

PoS (Proof-of-Stake) — A consensus method securing a blockchain through economic stake rather than energy-intensive computation.

Request for Interpretation (RFI) — The EAC’s formal mechanism by which a manufacturer or test laboratory obtains a definitive reading of a VVSG requirement (Testing and Certification Program Manual v3.0, §9); decisions are published.

RLA (Risk-Limiting Audit) — A statistical audit of paper ballots rigorous enough to confirm, to a chosen confidence level, that the reported winner won.

Salt — A secret random value mixed into a fingerprint so that the fingerprint reveals nothing about the ballot. A fresh salt is made for each ballot and kept only in the Secure Node’s sealed audit log.

Secure Node — A hardened polling facility with no network connection and electromagnetic shielding, where every ballot is printed, checked and filed.

SHA-256 — A NIST-approved cryptographic hash function (FIPS 180-4), used for every fingerprint the voting system computes.

STRIDE — A threat-modeling taxonomy: Spoofing, Tampering, Repudiation, Information disclosure, Denial of service, Elevation of privilege.

TEMPEST — The U.S. government’s name for the study and control of unintended electromagnetic emanations from equipment; facility countermeasures are assessed by a Certified TEMPEST Technical Authority.

Three-Key Rule — The requirement that every privileged action be authorized by three independent institutions at once (Section VI).

Title 32 — The part of federal law under which National Guard units serve in state-controlled status, where the Posse Comitatus Act does not apply.

Tripartite Oversight Triad — The three independent institutions that each hold one of the three keys (Section VI).

UOCAVA — Uniformed and Overseas Citizens Absentee Voting Act — the federal law governing voting by military and overseas citizens.

Validation index — A random eight-digit number printed on each ballot in the OCR-B typeface and read by optical character recognition. It identifies the ballot for filing and reconciliation, contains no vote, and cannot be linked to the voter (Section 8.1).

Validator script — An on-chain rule that decides whether an output may be spent. Here it allows a Vote Coin only to be burned — on a Secure Node’s signed record that the ballot was cast, or in the three-key close-out after certification (Section 14.1).

Vote Coin — A single-use, non-transferable token representing one voter’s right to one ballot in one election; minted before the election and burned after the ballot is cast (Phases I–II); from Phase III the same right is recorded as a spent credential tag in each anchored batch (Section 12.1).

Vote package — The digital record of one ballot — the record R the tablet produced (Section 8.1) — kept encrypted inside the Secure Node for reconciliation and audit, and never published or anchored.

VSTL — Voting System Test Laboratory — an EAC-accredited independent lab that tests voting systems against federal standards.

VVPAT — Voter-Verifiable Paper Audit Trail — a paper record the voter personally inspects; required by many states and by the draft VVSG 2.1.

VVSG 2.0 — Voluntary Voting System Guidelines 2.0 — the federal benchmark for voting systems, adopted by the EAC in 2021.

zk-SNARK / ZKP — Zero-knowledge proof: a cryptographic method of proving a statement true (e.g., “this voter is eligible”) without revealing anything else (e.g., who the voter is or how they voted).

FIGURE 27 · APPENDICES · PART EIGHT

The sample batch, in full

Eight sample ballots, their fingerprints, and the tree above them

Sample Node 01 · batch 0003 · 09:00–10:00 · 8 ballots — a sample: a real batch holds at least 100

#	INDEX	CHOICES (CONTEST 1, 2; MEASURE 3)	Fingerprint = SHA-256(R s), in full
1	 3618 3202	C · B · YES	55eaae1d b87959fa c20311ad fb5e9290 90aa350f aae7cb99 b6f7831a 5a24cdea
2	 3090 4408	B · A · YES	5ad00c95 8d482c2c a26db7fb a6ee5229 05c43491 53c0fdfa 5904f11b 863b679a
3	 7702 8142	A · B · YES	756f385f 9db9c96c c55af062 132aa543 41dbb7d2 642f3e0e ae6a7c13 6fdbba4b8
4	 0515 0293	A · A · NO	880d738d 9f978125 7ab513b3 12e5581f 71419543 b9c74f7e 5d6b7634 08e537d6
5	 5307 1166	A · B · YES	a1a467b8 8204e2b9 860c5e49 39851a17 32e2106f cc1c913c d1dfe336 2357600b
6	 9984 2665	A · B · YES	b245d512 657427cb d57920b9 20dcd7bf 26fee31f d56f0543 cb8b07df 1929f4df
7	 4199 9390	A · B · NO	d44e41fa 67da1fb0 5f33b7d6 644af88c 017fe74f e78c2e47 0bcacdc5 c73aec54
8	 4308 5720	C · A · NO SELECTION	f8bbfe7b db716361 d64920b2 a67cf78b acd1449e eb4a66ce fce0a293 cf3dbb10
Level 1	43e553d9 c8b8b6d0 d13516a0 881ac392		
Level 2	b83a4efc 00b17d9f		
Root	e2fe3a46 a74352ad 0ebd28c3 95758307 45cb2b80 5b4acfae 00c394f1 f8d72a75		

Choices are written A, B, C, YES, NO or NO SELECTION; in the record R each is spelled in full ("CHOICE A"), one per line.
Salts and the byte-for-byte records are in Figures: Worked Examples; these are sample ballots, not real votes.

8 → 1

8 fingerprints, sorted ascending
→ 8 leaves → 4 → 2 → 1 root (RFC 9162)
0293 is 4 of 8 by fingerprint, 1 of 8 by index

*Every fingerprint in the figures
starts on this page.*

The frozen sample batch behind the cover and every worked figure. The rows are in leaf order, sorted by fingerprint, so the tree can be rebuilt from this page alone.

Check it: hash 0x00 || each fingerprint for the leaves, pair them upward with 0x01, and the last pair gives e2fe3a46...f8d72a75.

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